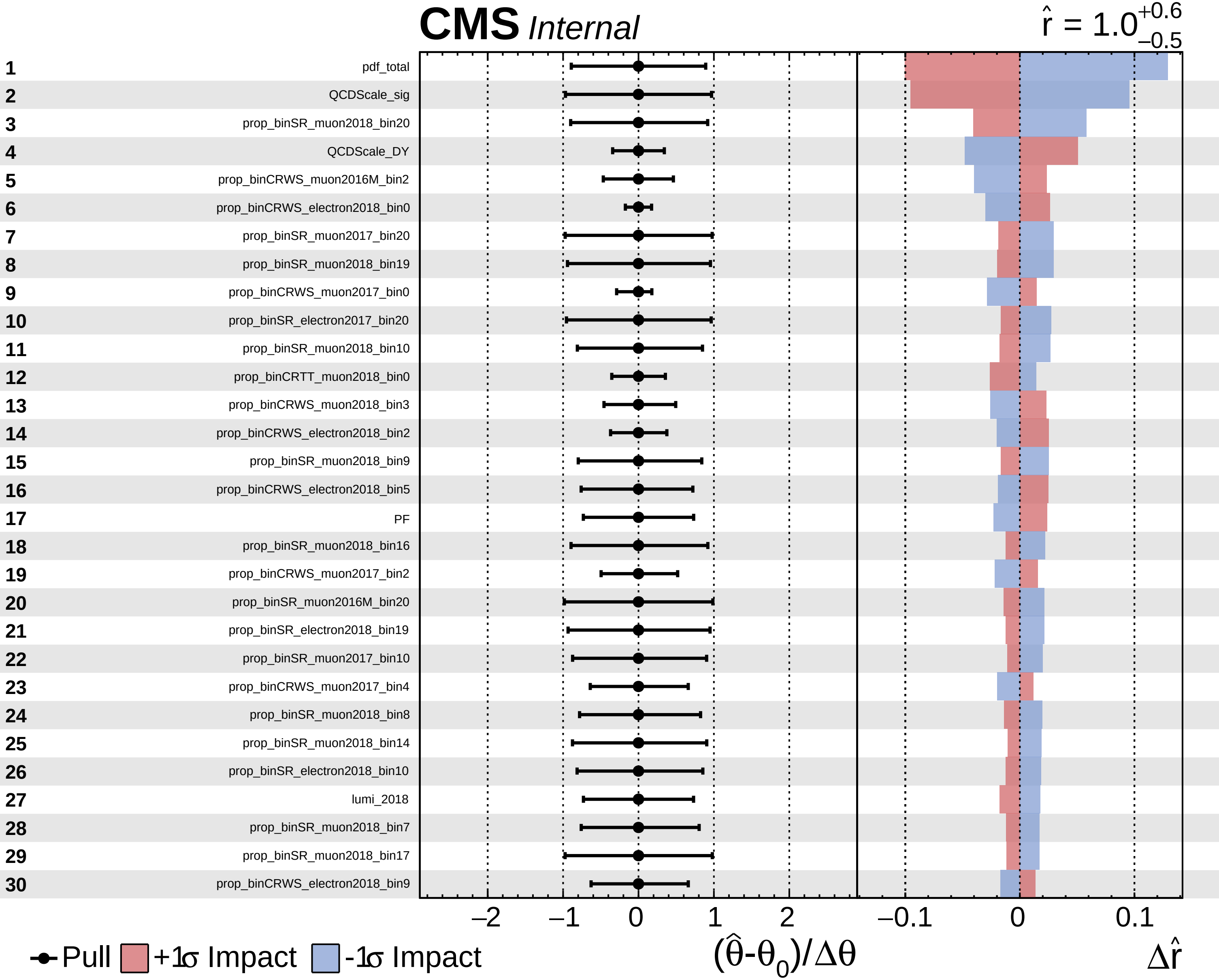
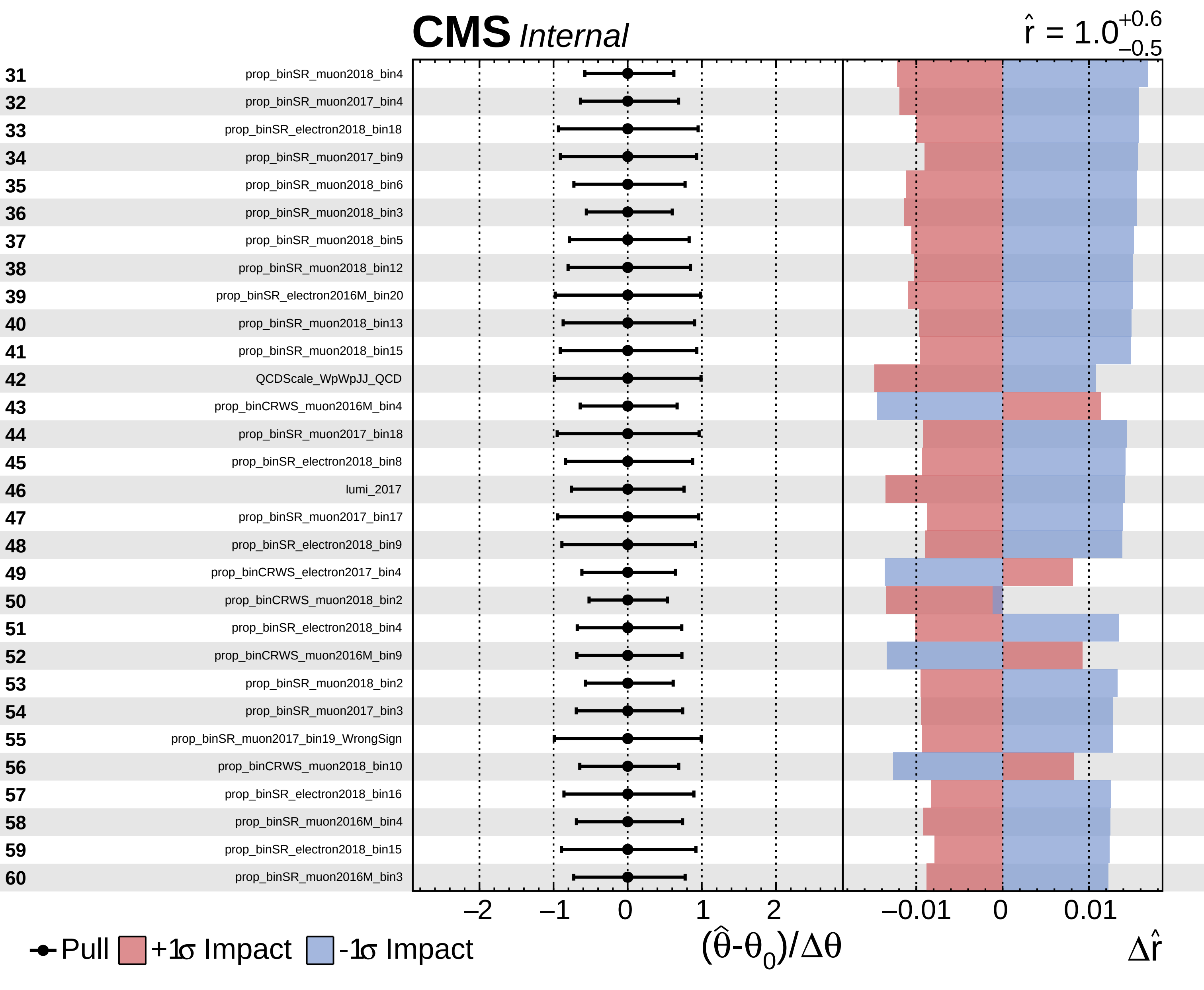
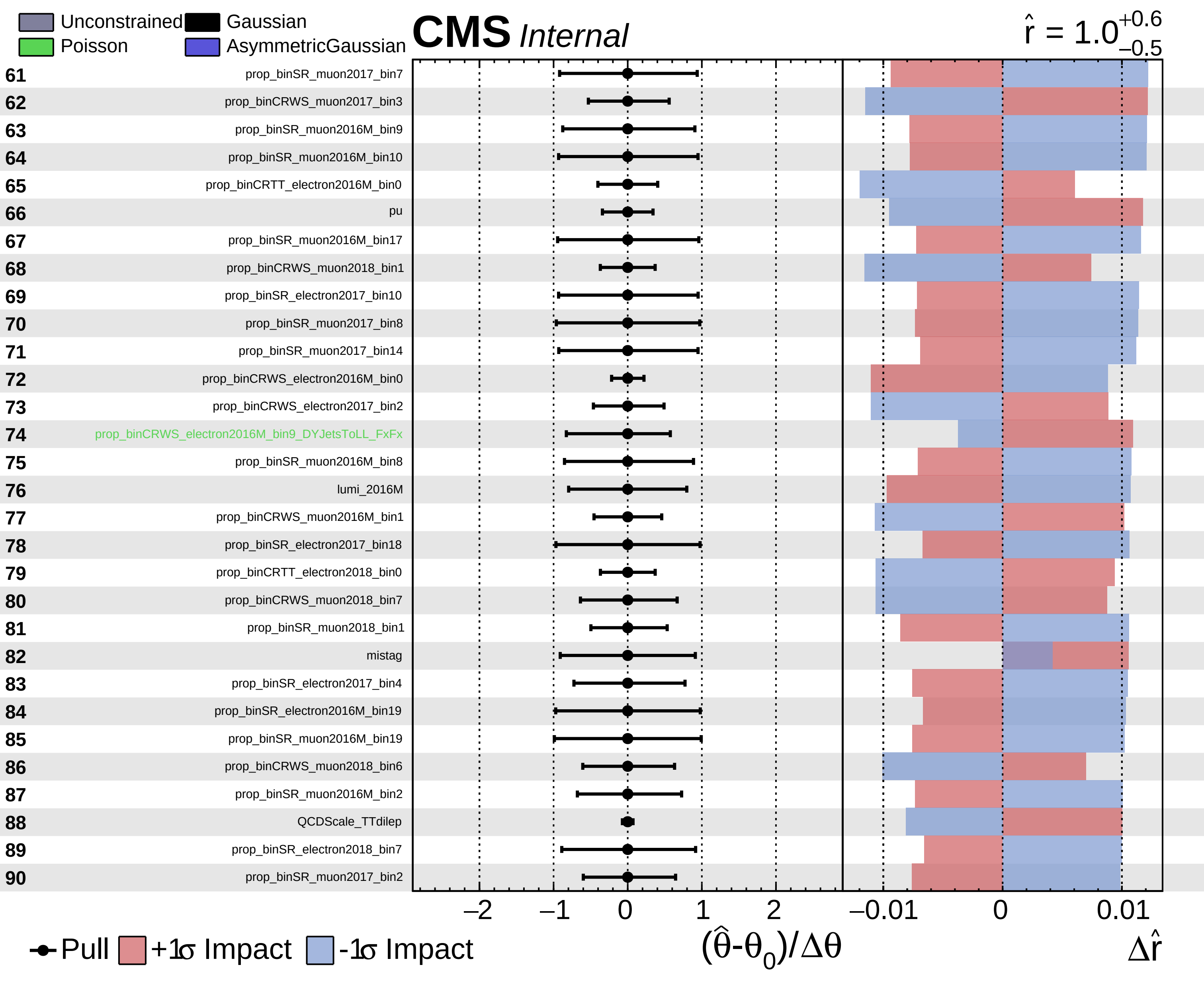
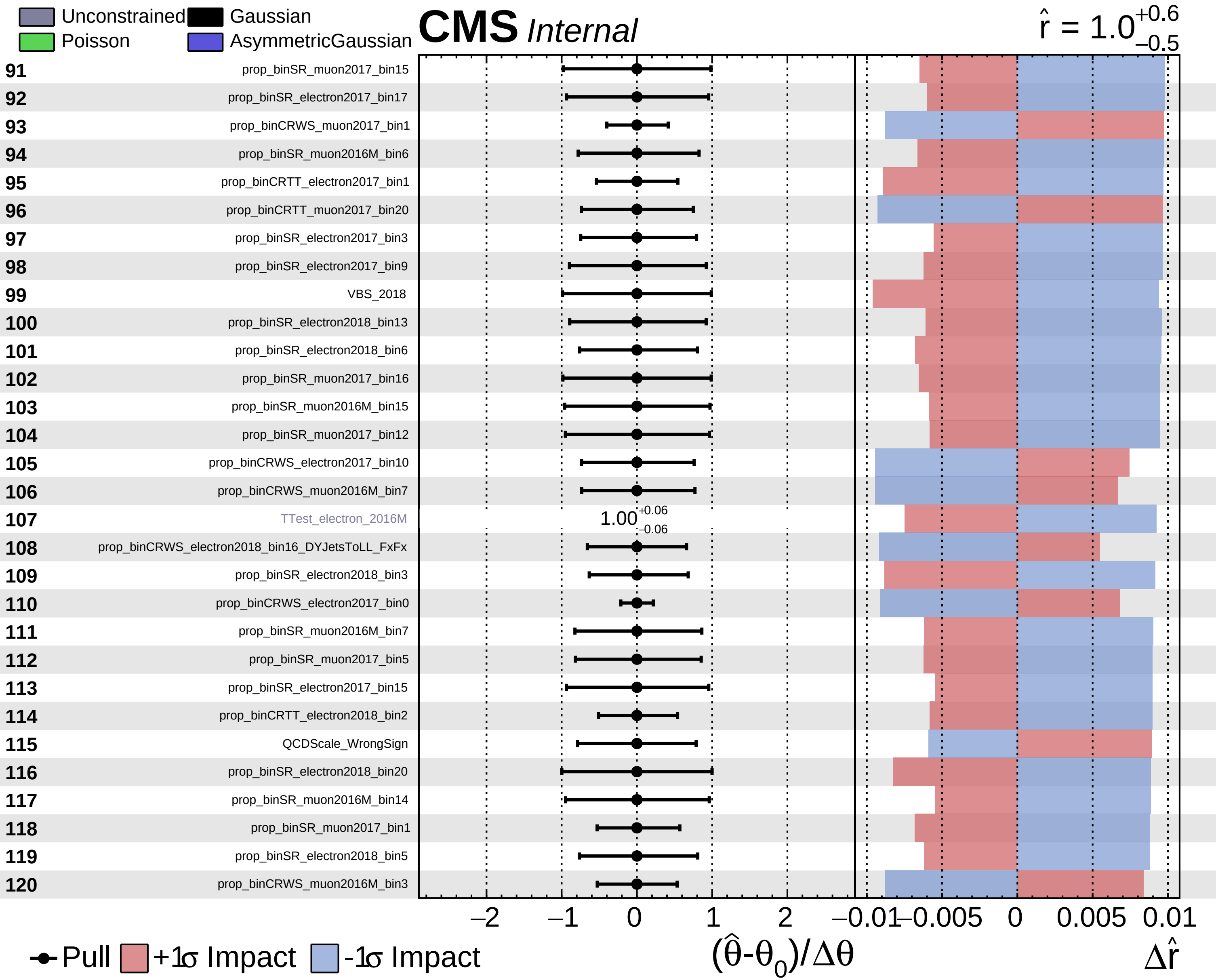


CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
1 pdf_total
2 QCDScale_sig
3 prop_binSR_muon2018_bin20
4 QCDScale_DY
5 prop_binCRWS_muon2016M_bin2
6 prop_binCRWS_electron2018_bin0
7 prop_binSR_muon2017_bin20
8 prop_binSR_muon2018_bin19
9 prop_binCRWS_muon2017_bin0
10 prop_binSR_electron2017_bin20
11 prop_binSR_muon2018_bin10
12 prop_binCRTT_muon2018_bin0
13 prop_binCRWS_muon2018_bin3
14 prop_binCRWS_electron2018_bin2
15 prop_binSR_muon2018_bin9
16 prop_binCRWS_electron2018_bin5
17 PF
18 prop_binSR_muon2018_bin16
19 prop_binCRWS_muon2017_bin2
20 prop_binSR_muon2016M_bin20
21 prop_binSR_electron2018_bin19
22 prop_binSR_muon2017_bin10
23 prop_binCRWS_muon2017_bin4
24 prop_binSR_muon2018_bin8
25 prop_binSR_muon2018_bin14
26 prop_binSR_electron2018_bin10
27 lumi_2018
28 prop_binSR_muon2018_bin7
29 prop_binSR_muon2018_bin17
30 prop_binCRWS_electron2018_bin9
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

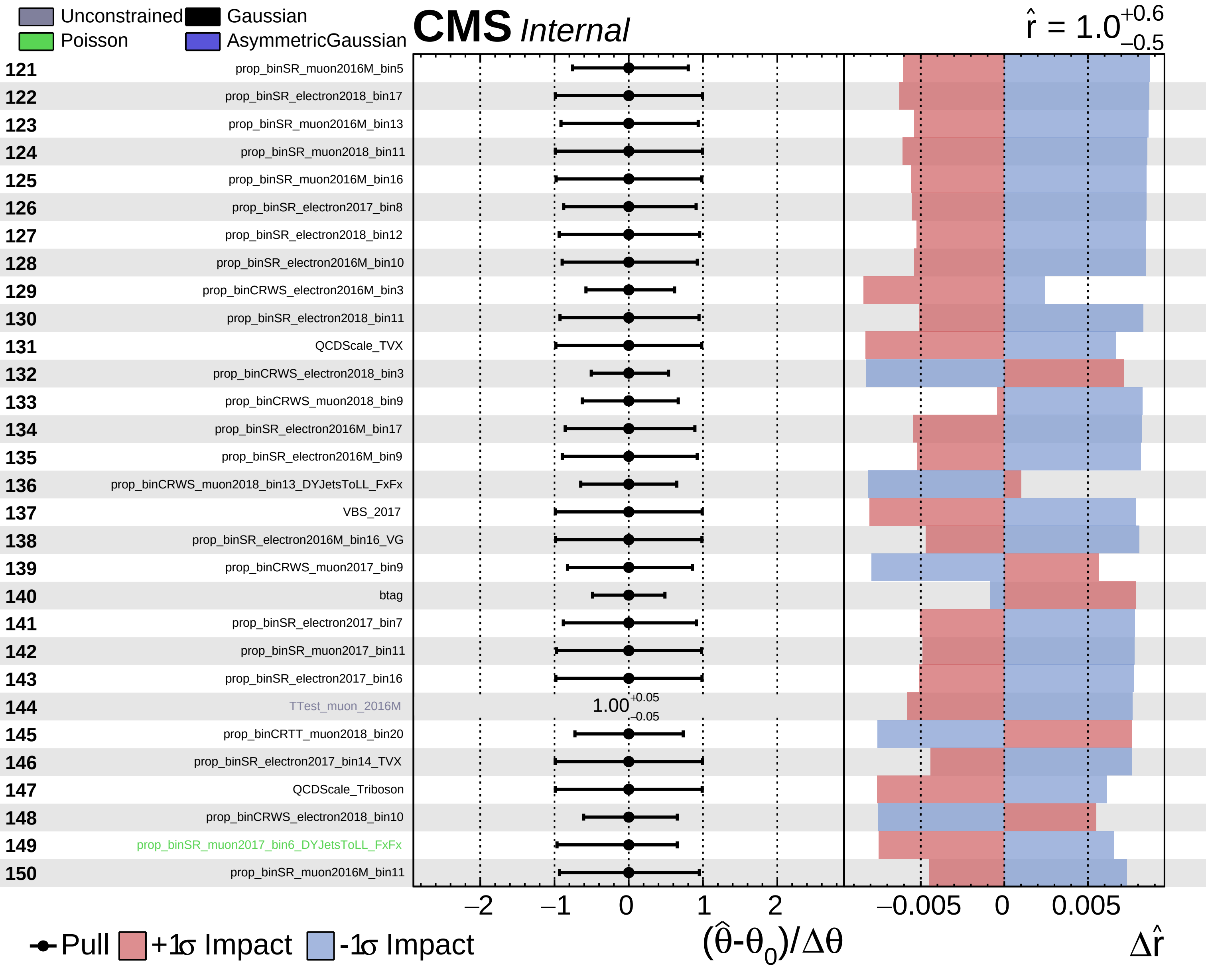

CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
31 prop_binSR_muon2018_bin4
32 prop_binSR_muon2017_bin4
33 prop_binSR_electron2018_bin18
34 prop_binSR_muon2017_bin9
35 prop_binSR_muon2018_bin6
36 prop_binSR_muon2018_bin3
37 prop_binSR_muon2018_bin5
38 prop_binSR_muon2018_bin12
39 prop_binSR_electron2016M_bin20
40 prop_binSR_muon2018_bin13
41 prop_binSR_muon2018_bin15
42 QCDScale_WpWpJJ_QCD
43 prop_binCRWS_muon2016M_bin4
44 prop_binSR_muon2017_bin18
45 prop_binSR_electron2018_bin8
46 lumi_2017
47 prop_binSR_muon2017_bin17
48 prop_binSR_electron2018_bin9
49 prop_binCRWS_electron2017_bin4
50 prop_binCRWS_muon2018_bin2
51 prop_binSR_electron2018_bin4
52 prop_binCRWS_muon2016M_bin9
53 prop_binSR_muon2018_bin2
54 prop_binSR_muon2017_bin3
55 prop_binSR_muon2017_bin19_WrongSign
56 prop_binCRWS_muon2018_bin10
57 prop_binSR_electron2018_bin16
58 prop_binSR_muon2016M_bin4
59 prop_binSR_electron2018_bin15
60 prop_binSR_muon2016M_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



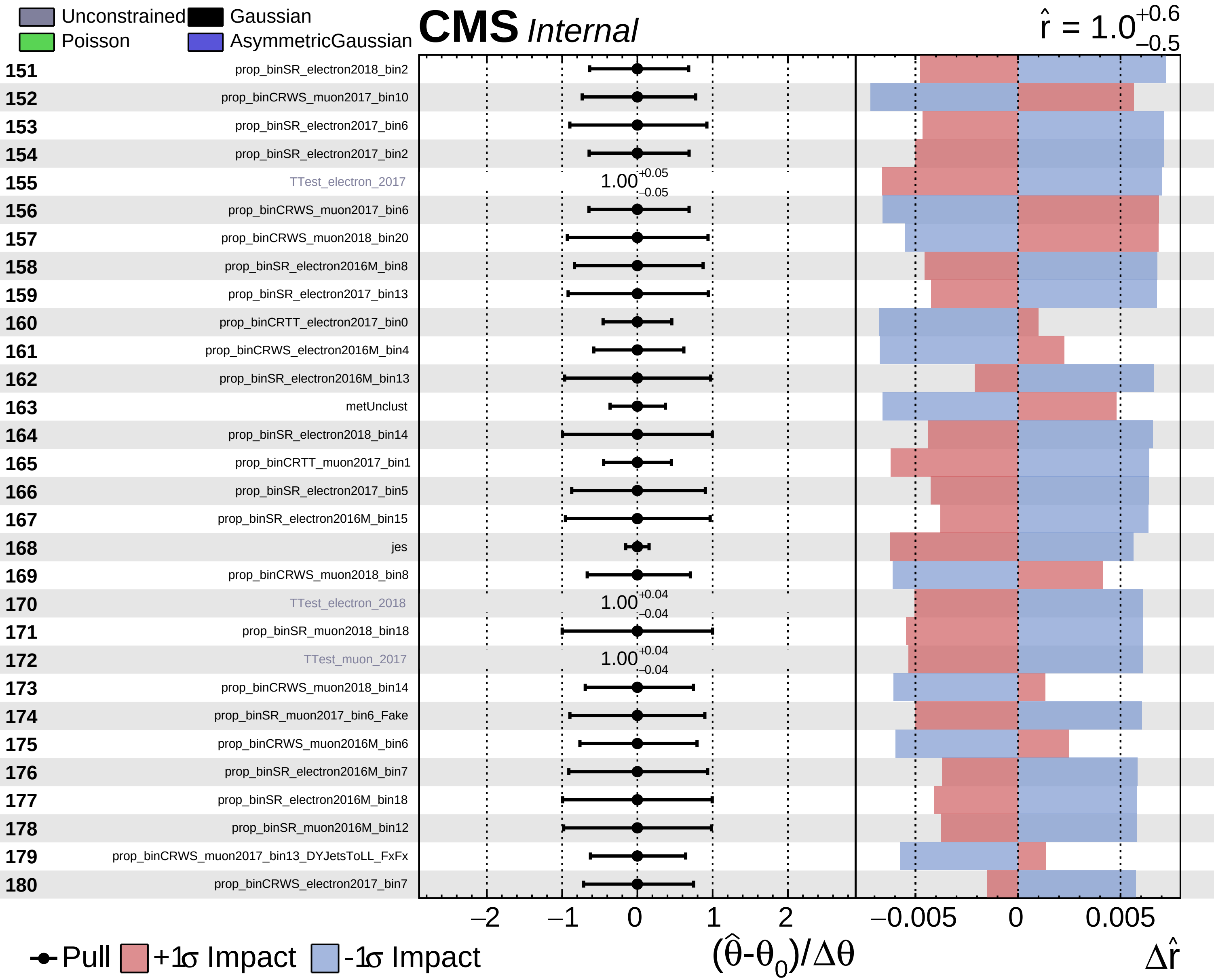
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
61 prop_binSR_muon2017_bin7
62 prop_binCRWS_muon2017_bin3
63 prop_binSR_muon2016M_bin9
64 prop_binSR_muon2016M_bin10
65 prop_binCRTT_electron2016M_bin0
66 pu
67 prop_binSR_muon2016M_bin17
68 prop_binCRWS_muon2018_bin1
69 prop_binSR_electron2017_bin10
70 prop_binSR_muon2017_bin8
71 prop_binSR_muon2017_bin14
72 prop_binCRWS_electron2016M_bin0
73 prop_binCRWS_electron2017_bin2
74 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
75 prop_binSR_muon2016M_bin8
76 lumi_2016M
77 prop_binCRWS_muon2016M_bin1
78 prop_binSR_electron2017_bin18
79 prop_binCRTT_electron2018_bin0
80 prop_binCRWS_muon2018_bin7
81 prop_binSR_muon2018_bin1
82 mistag
83 prop_binSR_electron2017_bin4
84 prop_binSR_electron2016M_bin19
85 prop_binSR_muon2016M_bin19
86 prop_binCRWS_muon2018_bin6
87 prop_binSR_muon2016M_bin2
88 QCDScale_TTdilep
89 prop_binSR_electron2018_bin7
90 prop_binSR_muon2017_bin2
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



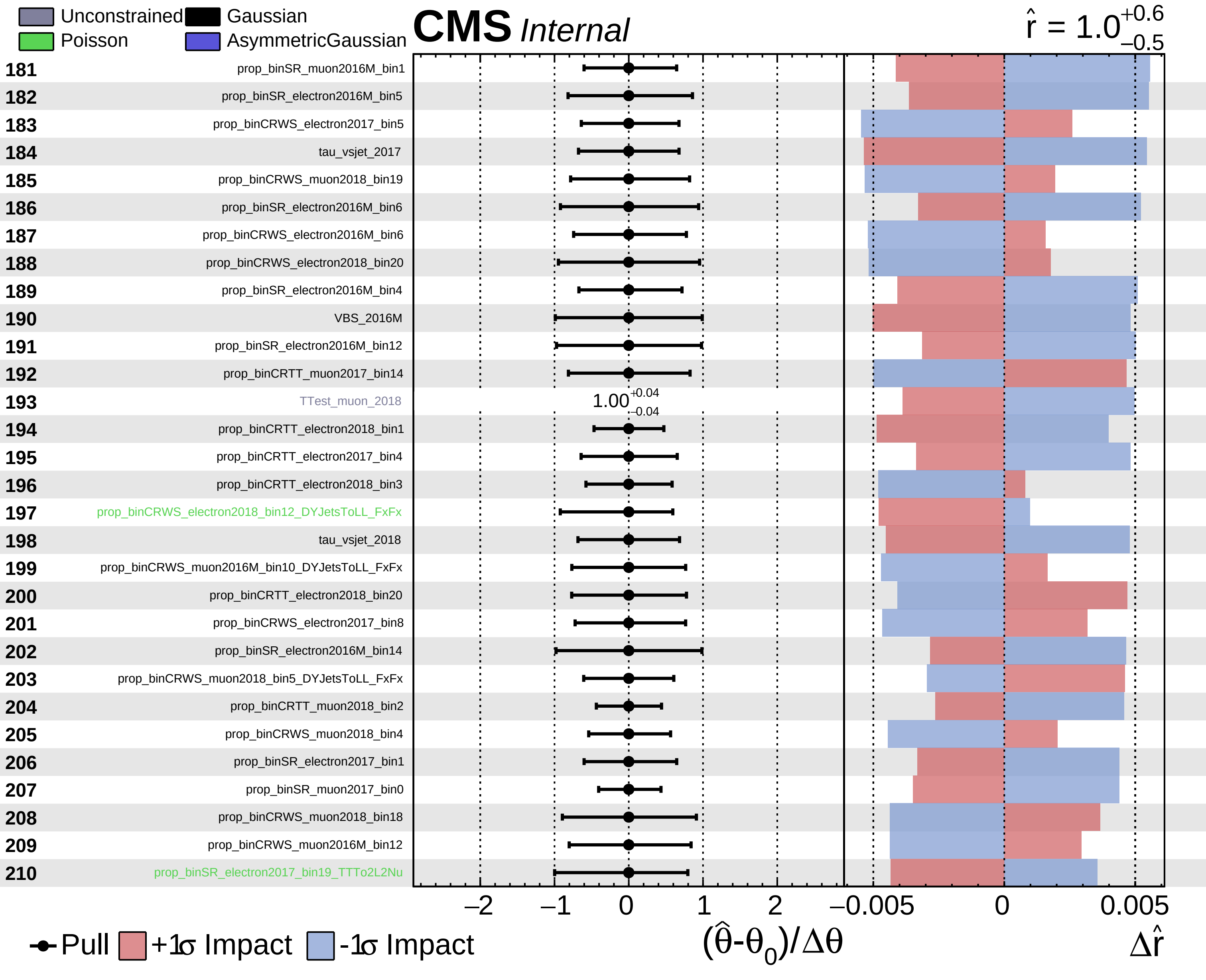
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
91 prop_binSR_muon2017_bin15
92 prop_binSR_electron2017_bin17
93 prop_binCRWS_muon2017_bin1
94 prop_binSR_muon2016M_bin6
95 prop_binCRTT_electron2017_bin1
96 prop_binCRTT_muon2017_bin20
97 prop_binSR_electron2017_bin3
98 prop_binSR_electron2017_bin9
99 VBS_2018
100 prop_binSR_electron2018_bin13
101 prop_binSR_electron2018_bin6
102 prop_binSR_muon2017_bin16
103 prop_binSR_muon2016M_bin15
104 prop_binSR_muon2017_bin12
105 prop_binCRWS_electron2017_bin10
106 prop_binCRWS_muon2016M_bin7
107 TTest_electron_2016M
1.00 +0.06 −0.06
108 prop_binCRWS_electron2018_bin16_DYJetsToLL_FxFx
109 prop_binSR_electron2018_bin3
110 prop_binCRWS_electron2017_bin0
111 prop_binSR_muon2016M_bin7
112 prop_binSR_muon2017_bin5
113 prop_binSR_electron2017_bin15
114 prop_binCRTT_electron2018_bin2
115 QCDScale_WrongSign
116 prop_binSR_electron2018_bin20
117 prop_binSR_muon2016M_bin14
118 prop_binSR_muon2017_bin1
119 prop_binSR_electron2018_bin5
120 prop_binCRWS_muon2016M_bin3
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



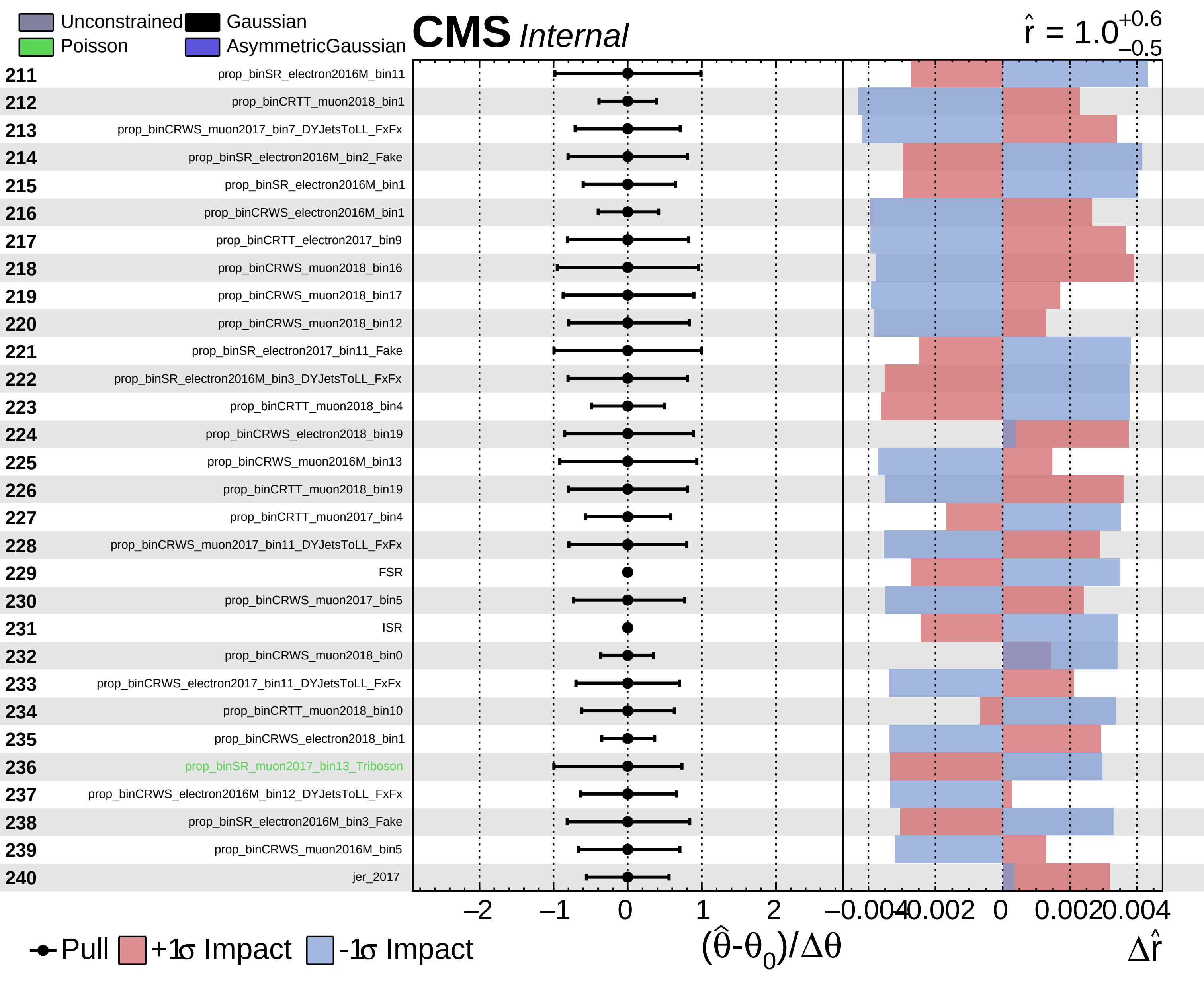
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
121 prop_binSR_muon2016M_bin5
122 prop_binSR_electron2018_bin17
123 prop_binSR_muon2016M_bin13
124 prop_binSR_muon2018_bin11
125 prop_binSR_muon2016M_bin16
126 prop_binSR_electron2017_bin8
127 prop_binSR_electron2018_bin12
128 prop_binSR_electron2016M_bin10
129 prop_binCRWS_electron2016M_bin3
130 prop_binSR_electron2018_bin11
131 QCDScale_TVX
132 prop_binCRWS_electron2018_bin3
133 prop_binCRWS_muon2018_bin9
134 prop_binSR_electron2016M_bin17
135 prop_binSR_electron2016M_bin9
136 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
137 VBS_2017
138 prop_binSR_electron2016M_bin16_VG
139 prop_binCRWS_muon2017_bin9
140 btag
141 prop_binSR_electron2017_bin7
142 prop_binSR_muon2017_bin11
143 prop_binSR_electron2017_bin16
144 TTest_muon_2016M
$1.00^{+0.05}_{-0.05}$
145 prop_binCRTT_muon2018_bin20
146 prop_binSR_electron2017_bin14_TVX
147 QCDScale_Triboson
148 prop_binCRWS_electron2018_bin10
149 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
150 prop_binSR_muon2016M_bin11
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



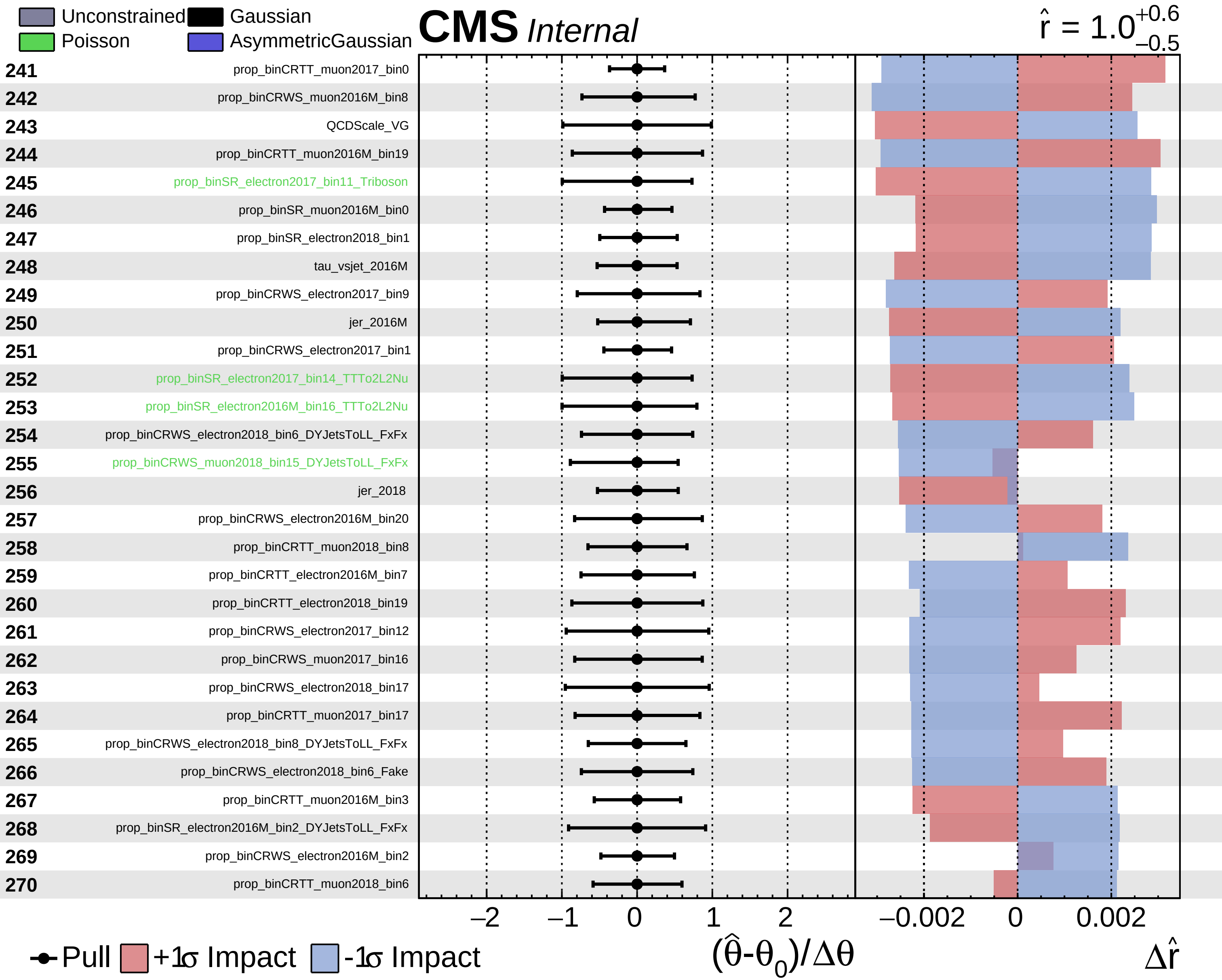
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
151 prop_binSR_electron2018_bin2
152 prop_binCRWS_muon2017_bin10
153 prop_binSR_electron2017_bin6
154 prop_binSR_electron2017_bin2
155 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
156 prop_binCRWS_muon2017_bin6
157 prop_binCRWS_muon2018_bin20
158 prop_binSR_electron2016M_bin8
159 prop_binSR_electron2017_bin13
160 prop_binCRTT_electron2017_bin0
161 prop_binCRWS_electron2016M_bin4
162 prop_binSR_electron2016M_bin13
163 metUnclust
164 prop_binSR_electron2018_bin14
165 prop_binCRTT_muon2017_bin1
166 prop_binSR_electron2017_bin5
167 prop_binSR_electron2016M_bin15
168 jes
169 prop_binCRWS_muon2018_bin8
170 TTest_electron_2018 $1.00^{+0.04}_{-0.04}$
171 prop_binSR_muon2018_bin18
172 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
173 prop_binCRWS_muon2018_bin14
174 prop_binSR_muon2017_bin6_Fake
175 prop_binCRWS_muon2016M_bin6
176 prop_binSR_electron2016M_bin7
177 prop_binSR_electron2016M_bin18
178 prop_binSR_muon2016M_bin12
179 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
180 prop_binCRWS_electron2017_bin7
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



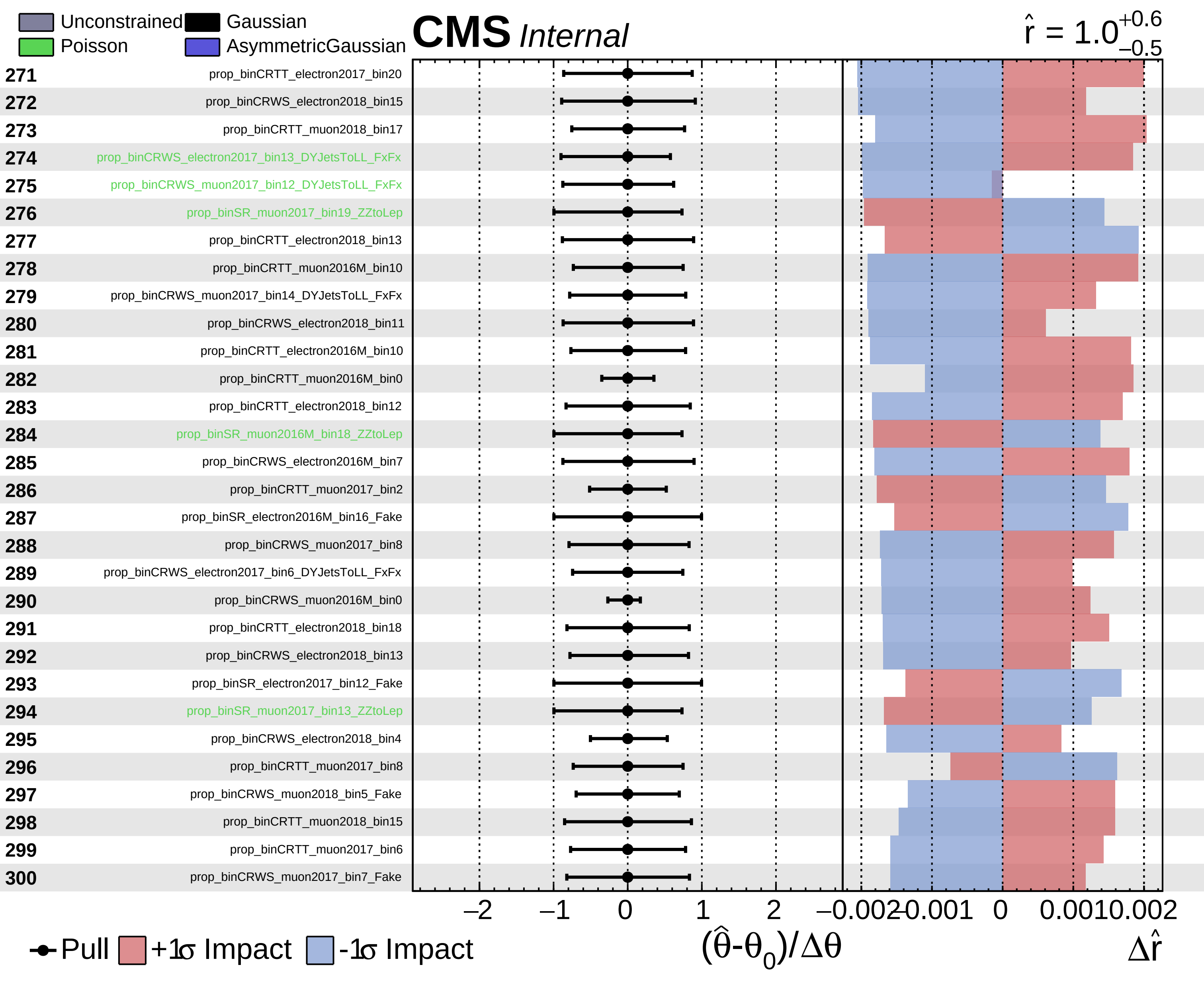
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
181 prop_binSR_muon2016M_bin1
182 prop_binSR_electron2016M_bin5
183 prop_binCRWS_electron2017_bin5
184 tau_vsjet_2017
185 prop_binCRWS_muon2018_bin19
186 prop_binSR_electron2016M_bin6
187 prop_binCRWS_electron2016M_bin6
188 prop_binCRWS_electron2018_bin20
189 prop_binSR_electron2016M_bin4
190 VBS_2016M
191 prop_binSR_electron2016M_bin12
192 prop_binCRTT_muon2017_bin14
193 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
194 prop_binCRTT_electron2018_bin1
195 prop_binCRTT_electron2017_bin4
196 prop_binCRTT_electron2018_bin3
197 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
198 tau_vsjet_2018
199 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
200 prop_binCRTT_electron2018_bin20
201 prop_binCRWS_electron2017_bin8
202 prop_binSR_electron2016M_bin14
203 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
204 prop_binCRTT_muon2018_bin2
205 prop_binCRWS_muon2018_bin4
206 prop_binSR_electron2017_bin1
207 prop_binSR_muon2017_bin0
208 prop_binCRWS_muon2018_bin18
209 prop_binCRWS_muon2016M_bin12
210 prop_binSR_electron2017_bin19_TTTo2L2Nu
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



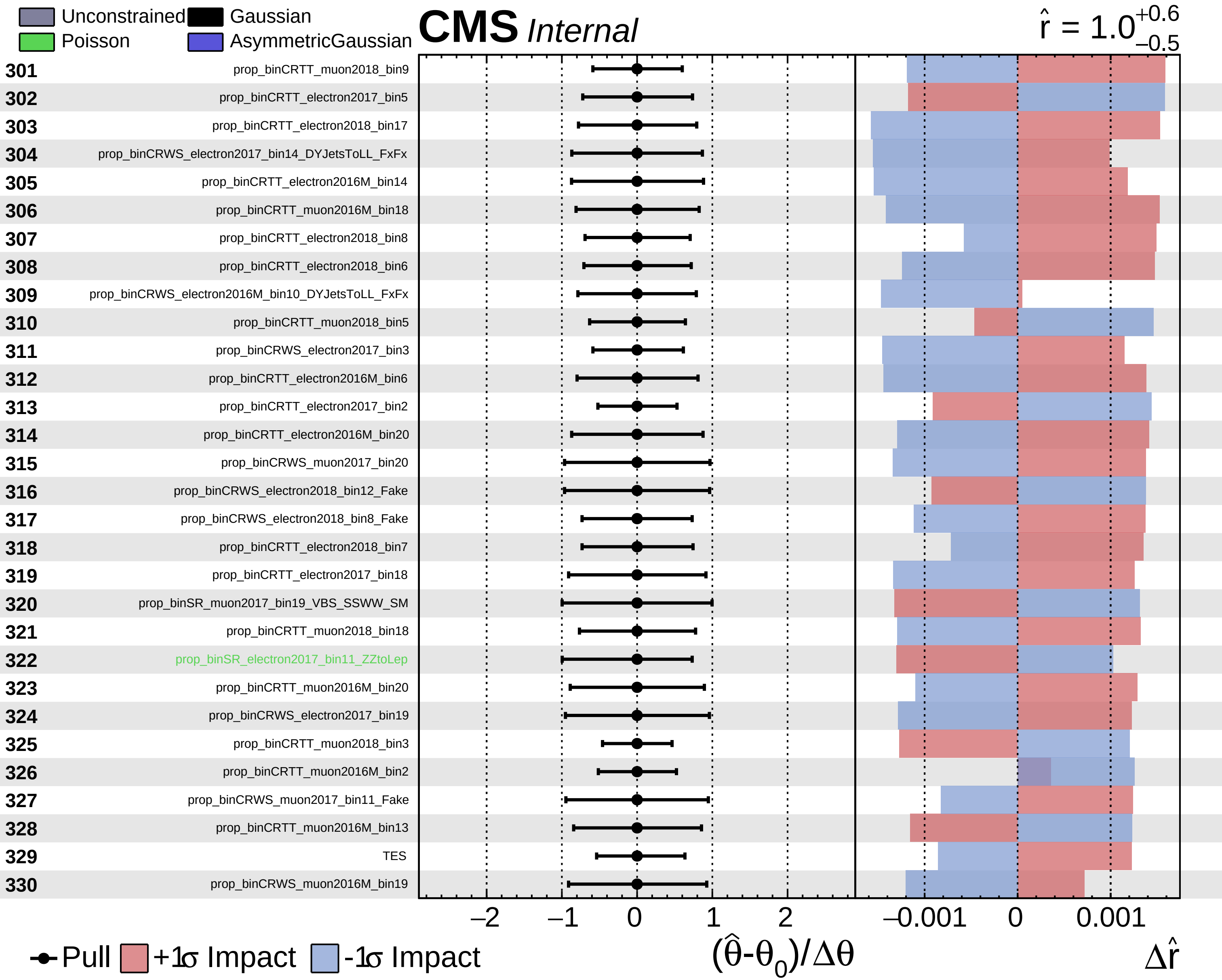
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
211 prop_binSR_electron2016M_bin11
212 prop_binCRTT_muon2018_bin1
213 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
214 prop_binSR_electron2016M_bin2_Fake
215 prop_binSR_electron2016M_bin1
216 prop_binCRWS_electron2016M_bin1
217 prop_binCRTT_electron2017_bin9
218 prop_binCRWS_muon2018_bin16
219 prop_binCRWS_muon2018_bin17
220 prop_binCRWS_muon2018_bin12
221 prop_binSR_electron2017_bin11_Fake
222 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
223 prop_binCRTT_muon2018_bin4
224 prop_binCRWS_electron2018_bin19
225 prop_binCRWS_muon2016M_bin13
226 prop_binCRTT_muon2018_bin19
227 prop_binCRTT_muon2017_bin4
228 prop_binCRWS_muon2017_bin11_DYJetsToLL_FxFx
229 FSR
230 prop_binCRWS_muon2017_bin5
231 ISR
232 prop_binCRWS_muon2018_bin0
233 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
234 prop_binCRTT_muon2018_bin10
235 prop_binCRWS_electron2018_bin1
236 prop_binSR_muon2017_bin13_Triboson
237 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
238 prop_binSR_electron2016M_bin3_Fake
239 prop_binCRWS_muon2016M_bin5
240 jer_2017
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



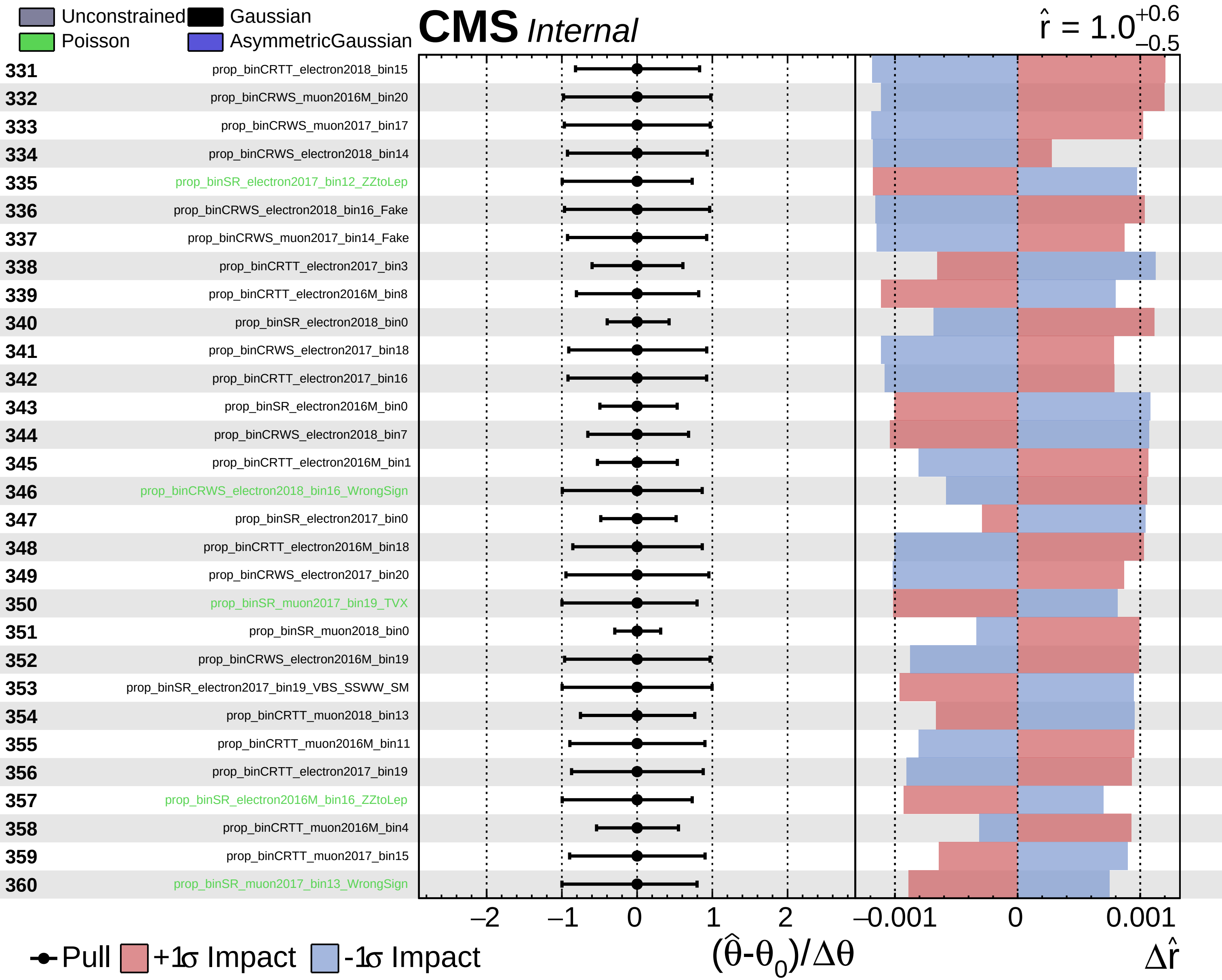
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
241 prop_binCRTT_muon2017_bin0
242 prop_binCRWS_muon2016M_bin8
243 QCDScale_VG
244 prop_binCRTT_muon2016M_bin19
245 prop_binSR_electron2017_bin11_Triboson
246 prop_binSR_muon2016M_bin0
247 prop_binSR_electron2018_bin1
248 tau_vsjet_2016M
249 prop_binCRWS_electron2017_bin9
250 jer_2016M
251 prop_binCRWS_electron2017_bin1
252 prop_binSR_electron2017_bin14_TTTo2L2Nu
253 prop_binSR_electron2016M_bin16_TTTo2L2Nu
254 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
255 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
256 jer_2018
257 prop_binCRWS_electron2016M_bin20
258 prop_binCRTT_muon2018_bin8
259 prop_binCRTT_electron2016M_bin7
260 prop_binCRTT_electron2018_bin19
261 prop_binCRWS_electron2017_bin12
262 prop_binCRWS_muon2017_bin16
263 prop_binCRWS_electron2018_bin17
264 prop_binCRTT_muon2017_bin17
265 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
266 prop_binCRWS_electron2018_bin6_Fake
267 prop_binCRTT_muon2016M_bin3
268 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
269 prop_binCRWS_electron2016M_bin2
270 prop_binCRTT_muon2018_bin6
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



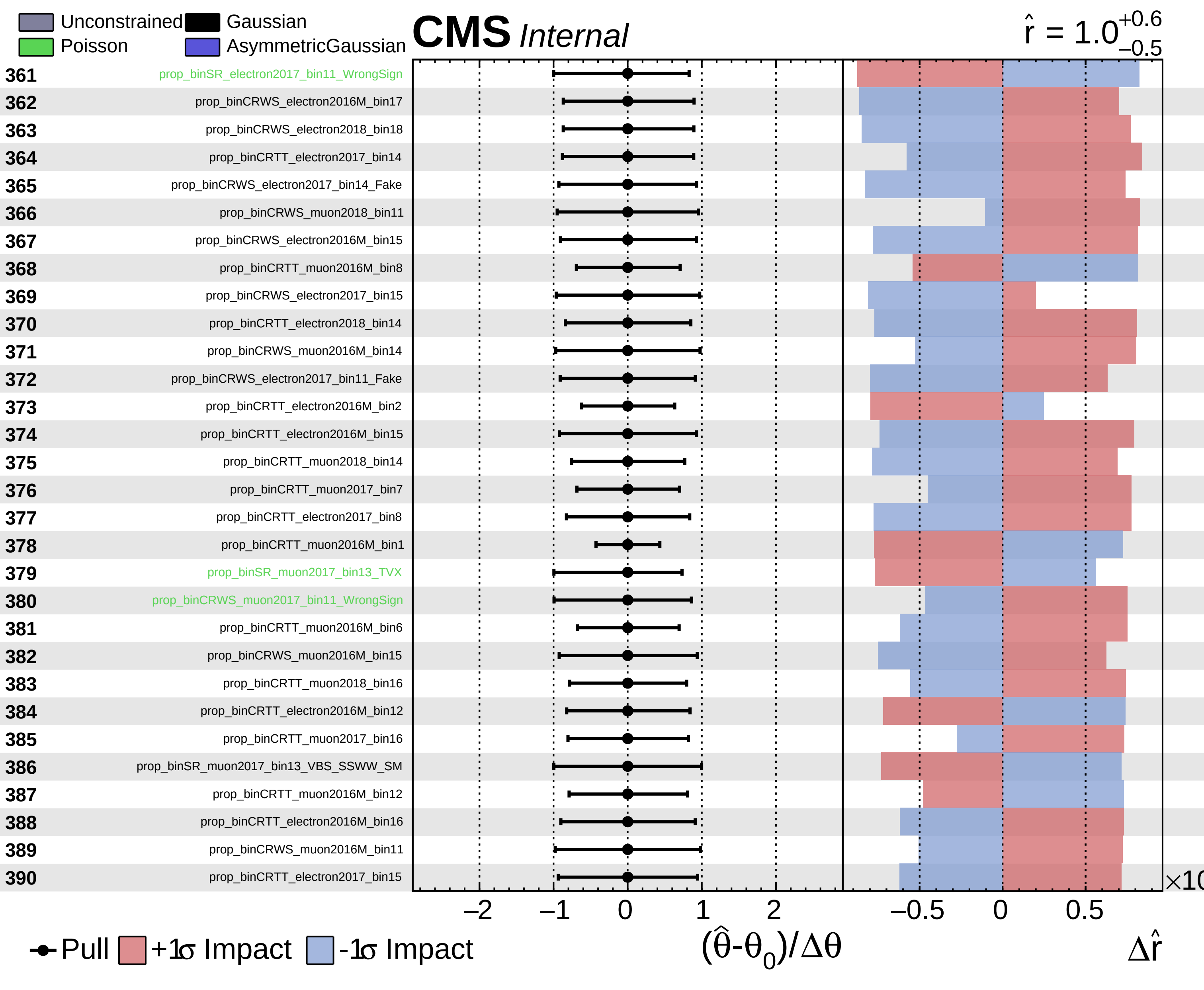
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
271 prop_binCRTT_electron2017_bin20
272 prop_binCRWS_electron2018_bin15
273 prop_binCRTT_muon2018_bin17
274 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
275 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
276 prop_binSR_muon2017_bin19_ZZtoLep
277 prop_binCRTT_electron2018_bin13
278 prop_binCRTT_muon2016M_bin10
279 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
280 prop_binCRWS_electron2018_bin11
281 prop_binCRTT_electron2016M_bin10
282 prop_binCRTT_muon2016M_bin0
283 prop_binCRTT_electron2018_bin12
284 prop_binSR_muon2016M_bin18_ZZtoLep
285 prop_binCRWS_electron2016M_bin7
286 prop_binCRTT_muon2017_bin2
287 prop_binSR_electron2016M_bin16_Fake
288 prop_binCRWS_muon2017_bin8
289 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
290 prop_binCRWS_muon2016M_bin0
291 prop_binCRTT_electron2018_bin18
292 prop_binCRWS_electron2018_bin13
293 prop_binSR_electron2017_bin12_Fake
294 prop_binSR_muon2017_bin13_ZZtoLep
295 prop_binCRWS_electron2018_bin4
296 prop_binCRTT_muon2017_bin8
297 prop_binCRWS_muon2018_bin5_Fake
298 prop_binCRTT_muon2018_bin15
299 prop_binCRTT_muon2017_bin6
300 prop_binCRWS_muon2017_bin7_Fake
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



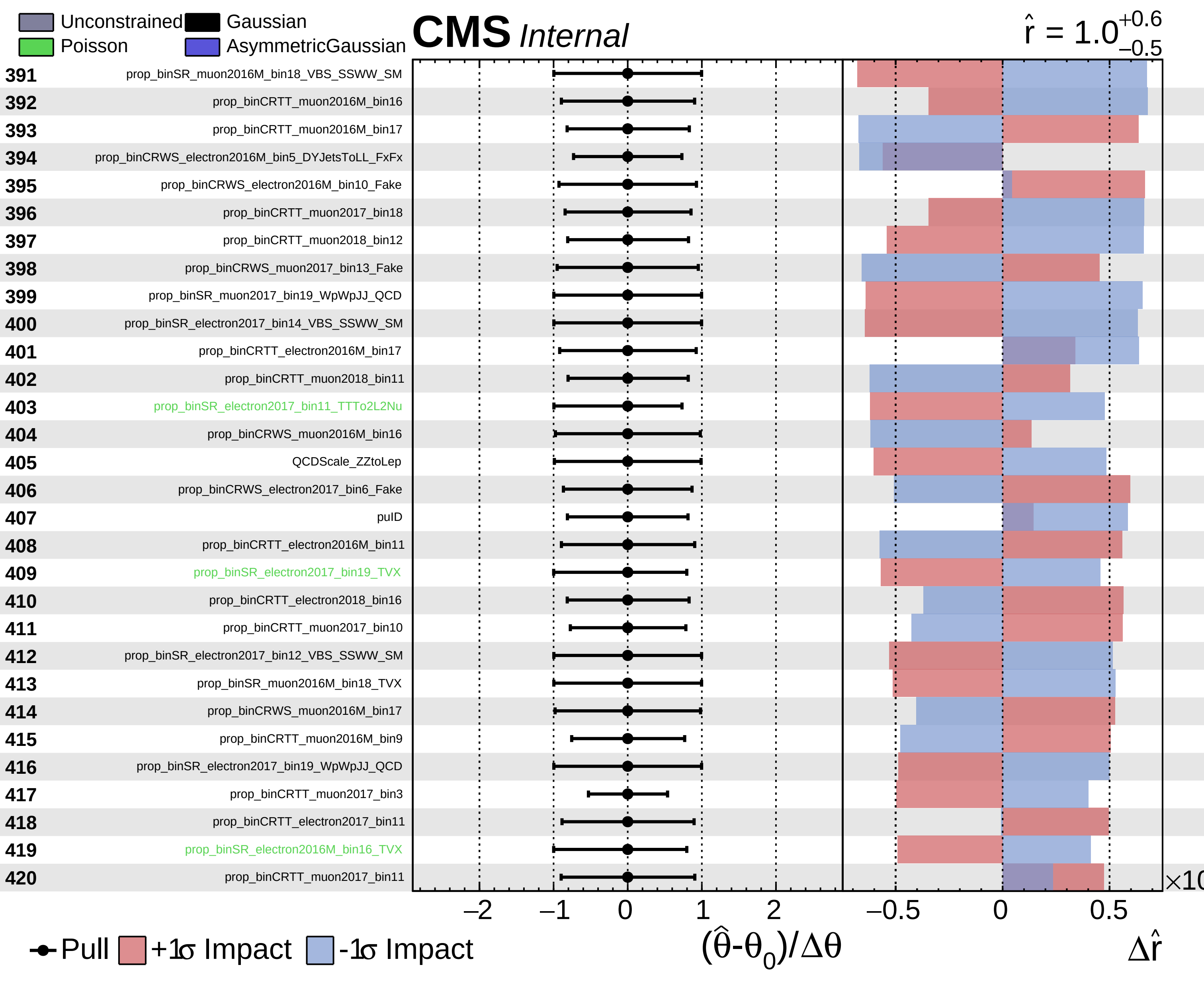
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
301 prop_binCRTT_muon2018_bin9
302 prop_binCRTT_electron2017_bin5
303 prop_binCRTT_electron2018_bin17
304 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
305 prop_binCRTT_electron2016M_bin14
306 prop_binCRTT_muon2016M_bin18
307 prop_binCRTT_electron2018_bin8
308 prop_binCRTT_electron2018_bin6
309 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
310 prop_binCRTT_muon2018_bin5
311 prop_binCRWS_electron2017_bin3
312 prop_binCRTT_electron2016M_bin6
313 prop_binCRTT_electron2017_bin2
314 prop_binCRTT_electron2016M_bin20
315 prop_binCRWS_muon2017_bin20
316 prop_binCRWS_electron2018_bin12_Fake
317 prop_binCRWS_electron2018_bin8_Fake
318 prop_binCRTT_electron2018_bin7
319 prop_binCRTT_electron2017_bin18
320 prop_binSR_muon2017_bin19_VBS_SSWW_SM
321 prop_binCRTT_muon2018_bin18
322 prop_binSR_electron2017_bin11_ZZtoLep
323 prop_binCRTT_muon2016M_bin20
324 prop_binCRWS_electron2017_bin19
325 prop_binCRTT_muon2018_bin3
326 prop_binCRTT_muon2016M_bin2
327 prop_binCRWS_muon2017_bin11_Fake
328 prop_binCRTT_muon2016M_bin13
329 TES
330 prop_binCRWS_muon2016M_bin19
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

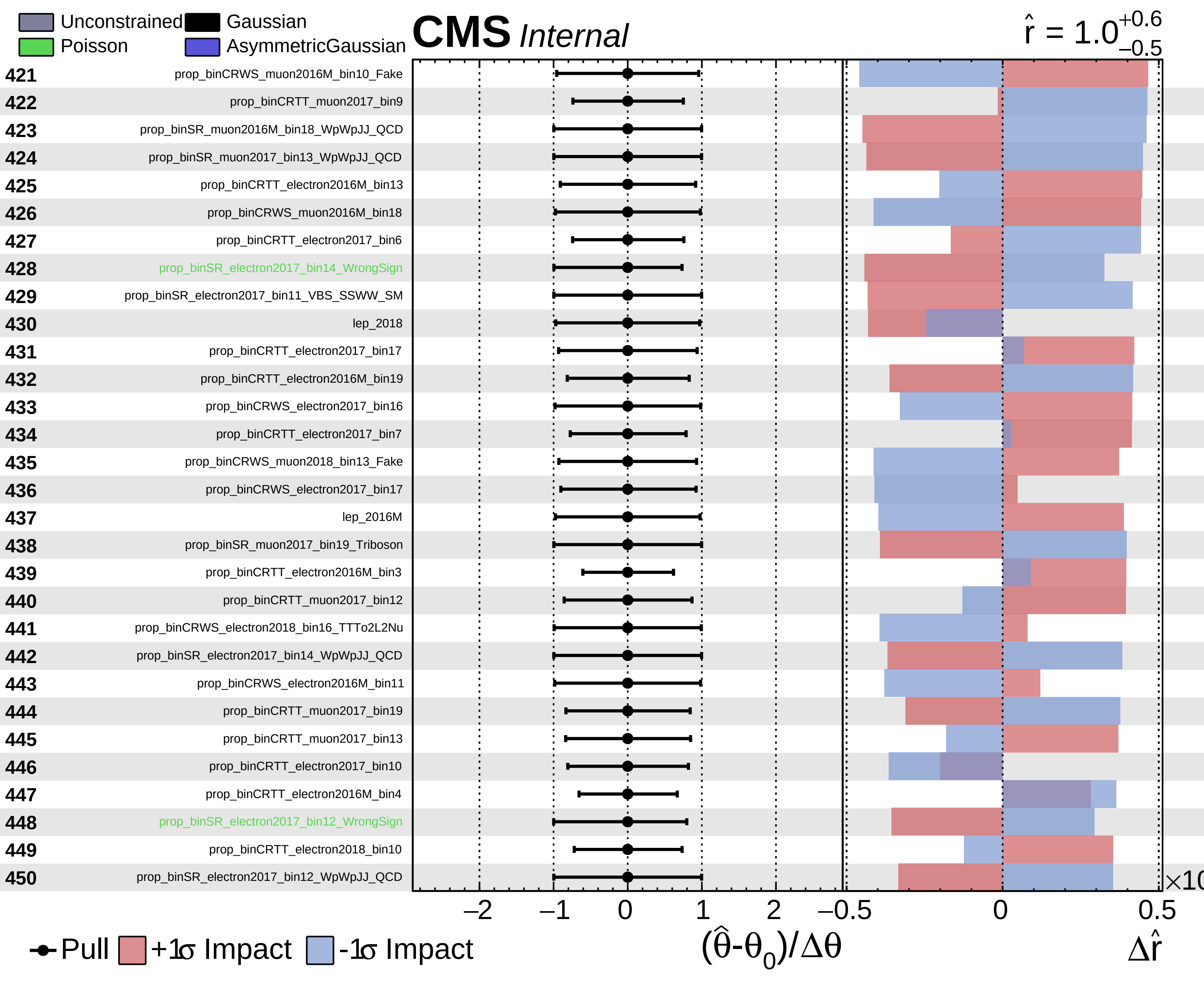


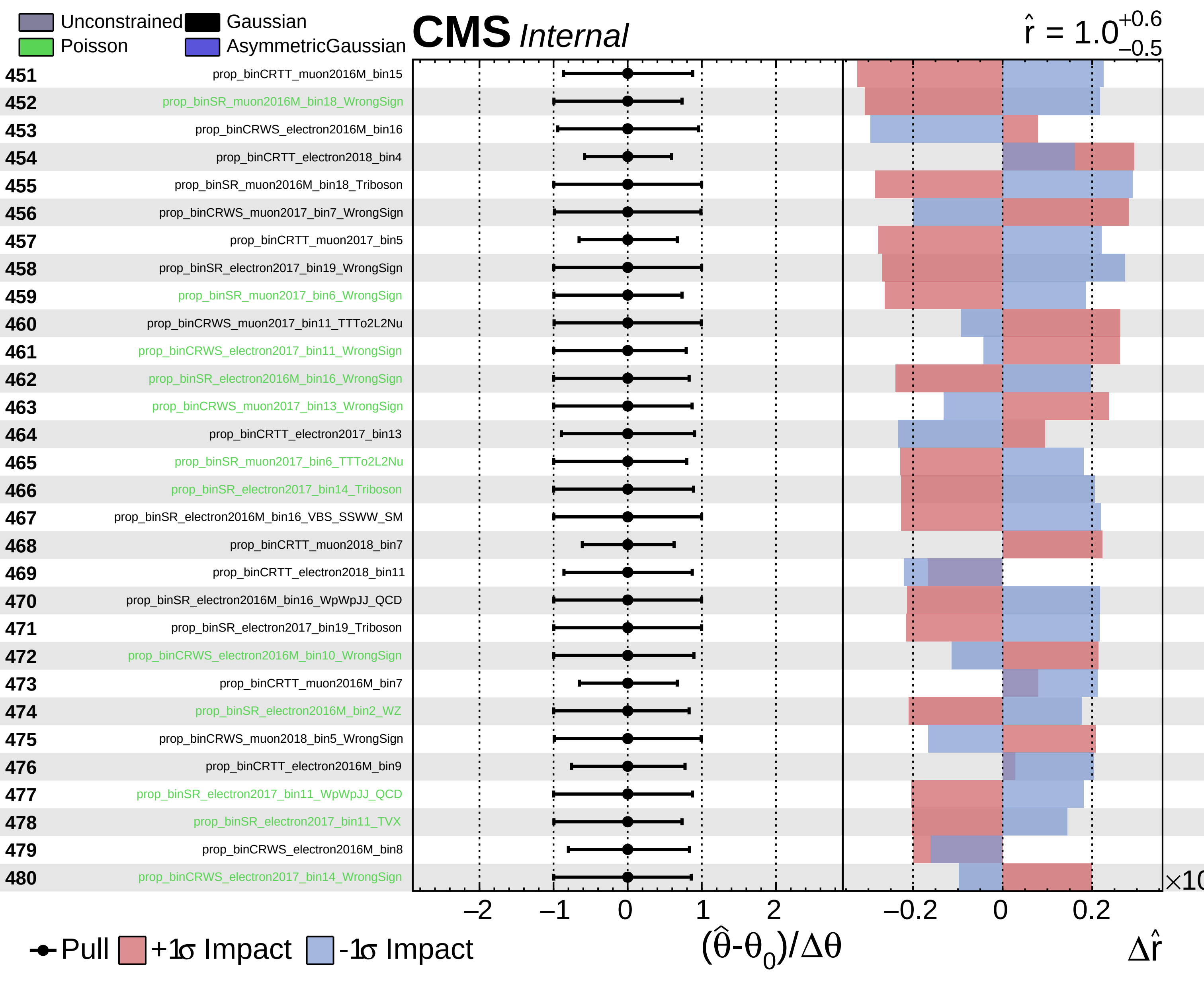
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
331 prop_binCRTT_electron2018_bin15
332 prop_binCRWS_muon2016M_bin20
333 prop_binCRWS_muon2017_bin17
334 prop_binCRWS_electron2018_bin14
335 prop_binSR_electron2017_bin12_ZZtoLep
336 prop_binCRWS_electron2018_bin16_Fake
337 prop_binCRWS_muon2017_bin14_Fake
338 prop_binCRTT_electron2017_bin3
339 prop_binCRTT_electron2016M_bin8
340 prop_binSR_electron2018_bin0
341 prop_binCRWS_electron2017_bin18
342 prop_binCRTT_electron2017_bin16
343 prop_binSR_electron2016M_bin0
344 prop_binCRWS_electron2018_bin7
345 prop_binCRTT_electron2016M_bin1
346 prop_binCRWS_electron2018_bin16_WrongSign
347 prop_binSR_electron2017_bin0
348 prop_binCRTT_electron2016M_bin18
349 prop_binCRWS_electron2017_bin20
350 prop_binSR_muon2017_bin19_TVX
351 prop_binSR_muon2018_bin0
352 prop_binCRWS_electron2016M_bin19
353 prop_binSR_electron2017_bin19_VBS_SSWW_SM
354 prop_binCRTT_muon2018_bin13
355 prop_binCRTT_muon2016M_bin11
356 prop_binCRTT_electron2017_bin19
357 prop_binSR_electron2016M_bin16_ZZtoLep
358 prop_binCRTT_muon2016M_bin4
359 prop_binCRTT_muon2017_bin15
360 prop_binSR_muon2017_bin13_WrongSign
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



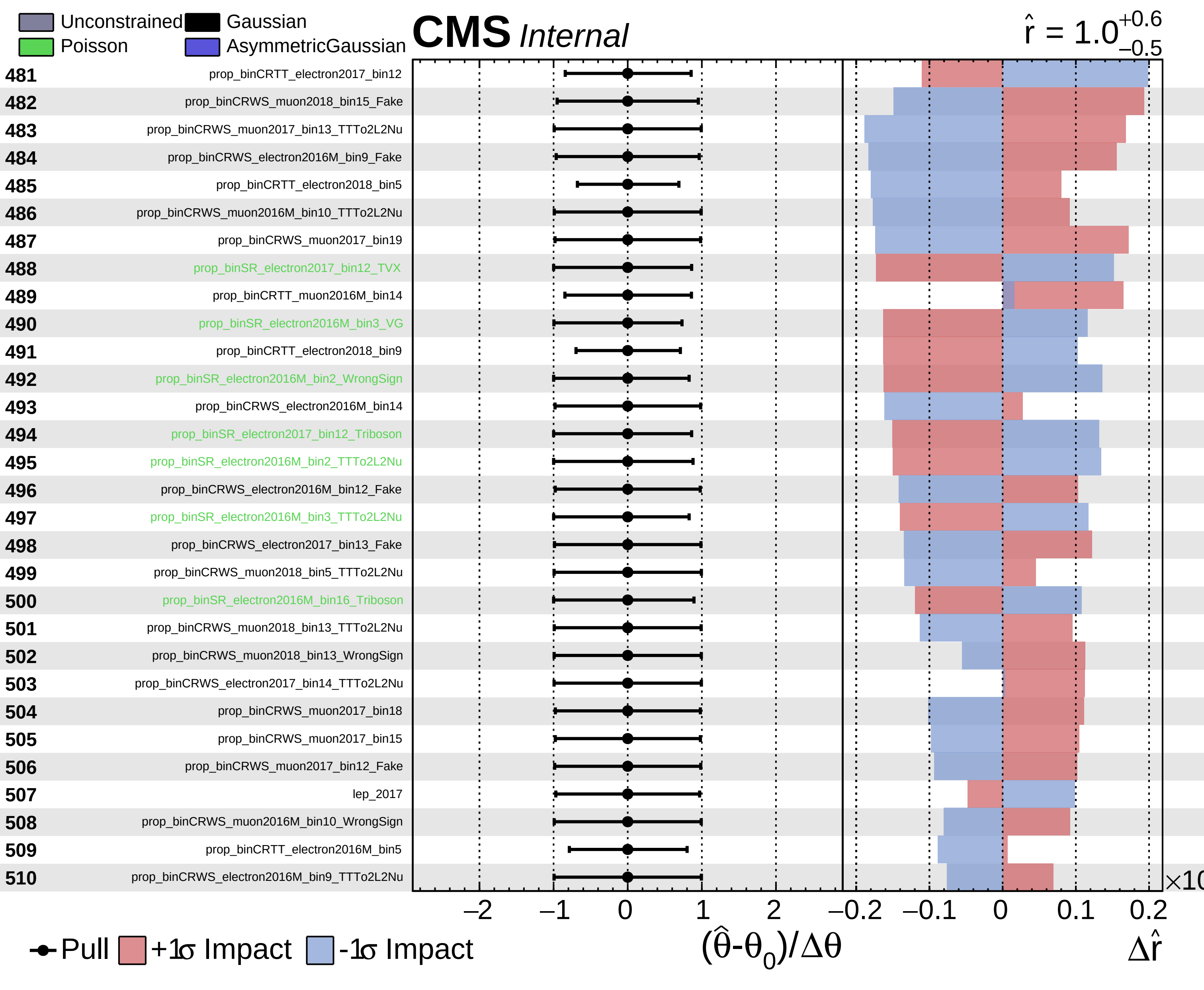
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
361 prop_binSR_electron2017_bin11_WrongSign
362 prop_binCRWS_electron2016M_bin17
363 prop_binCRWS_electron2018_bin18
364 prop_binCRTT_electron2017_bin14
365 prop_binCRWS_electron2017_bin14_Fake
366 prop_binCRWS_muon2018_bin11
367 prop_binCRWS_electron2016M_bin15
368 prop_binCRTT_muon2016M_bin8
369 prop_binCRWS_electron2017_bin15
370 prop_binCRTT_electron2018_bin14
371 prop_binCRWS_muon2016M_bin14
372 prop_binCRWS_electron2017_bin11_Fake
373 prop_binCRTT_electron2016M_bin2
374 prop_binCRTT_electron2016M_bin15
375 prop_binCRTT_muon2018_bin14
376 prop_binCRTT_muon2017_bin7
377 prop_binCRTT_electron2017_bin8
378 prop_binCRTT_muon2016M_bin1
379 prop_binSR_muon2017_bin13_TVX
380 prop_binCRWS_muon2017_bin11_WrongSign
381 prop_binCRTT_muon2016M_bin6
382 prop_binCRWS_muon2016M_bin15
383 prop_binCRTT_muon2018_bin16
384 prop_binCRTT_electron2016M_bin12
385 prop_binCRTT_muon2017_bin16
386 prop_binSR_muon2017_bin13_VBS_SSWW_SM
387 prop_binCRTT_muon2016M_bin12
388 prop_binCRTT_electron2016M_bin16
389 prop_binCRWS_muon2016M_bin11
390 prop_binCRTT_electron2017_bin15
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

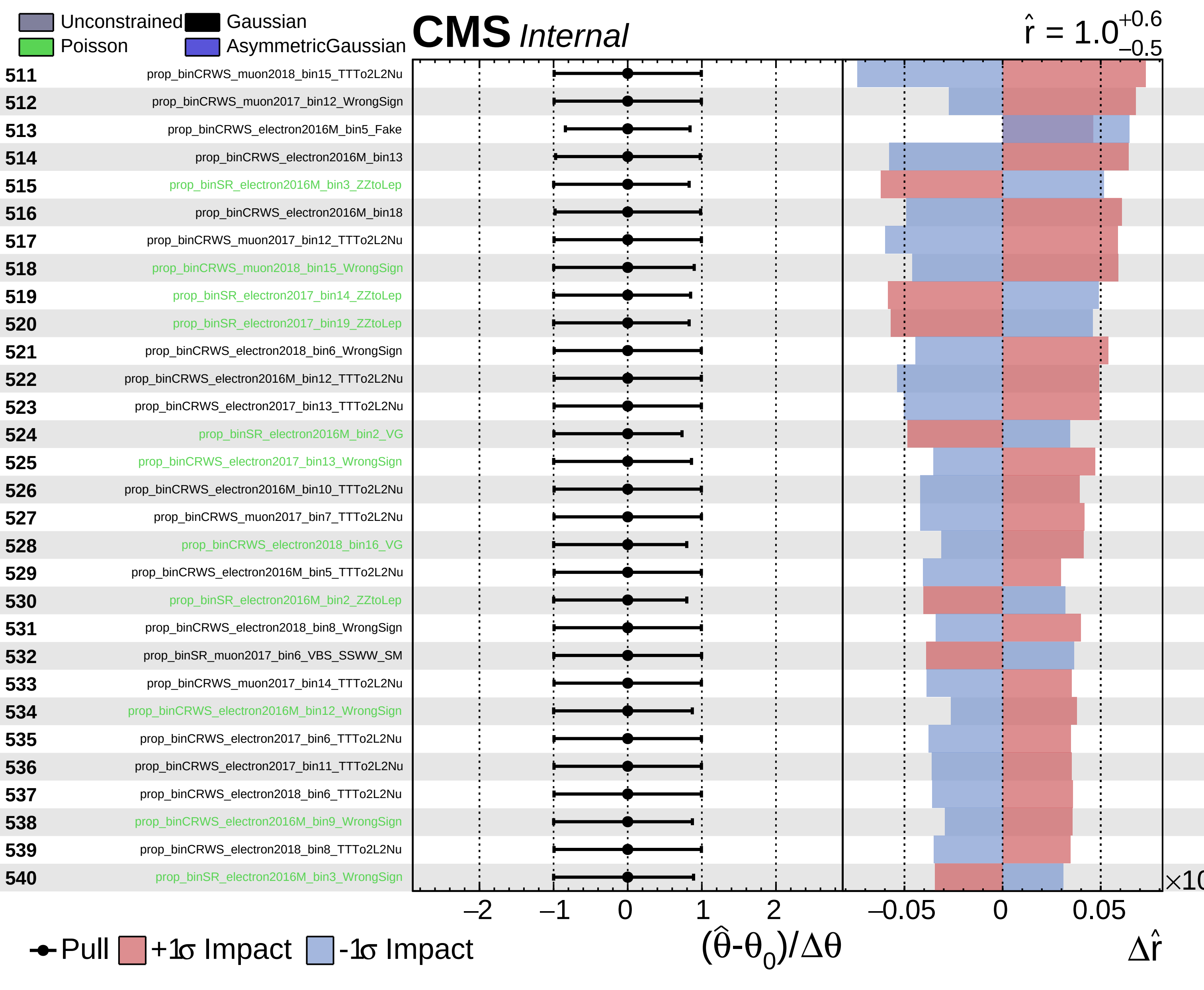

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
391 prop_binSR_muon2016M_bin18_VBS_SSWW_SM
392 prop_binCRTT_muon2016M_bin16
393 prop_binCRTT_muon2016M_bin17
394 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
395 prop_binCRWS_electron2016M_bin10_Fake
396 prop_binCRTT_muon2017_bin18
397 prop_binCRTT_muon2018_bin12
398 prop_binCRWS_muon2017_bin13_Fake
399 prop_binSR_muon2017_bin19_WpWpJJ_QCD
400 prop_binSR_electron2017_bin14_VBS_SSWW_SM
401 prop_binCRTT_electron2016M_bin17
402 prop_binCRTT_muon2018_bin11
403 prop_binSR_electron2017_bin11_TTTo2L2Nu
404 prop_binCRWS_muon2016M_bin16
405 QCDScale_ZZtoLep
406 prop_binCRWS_electron2017_bin6_Fake
407 puID
408 prop_binCRTT_electron2016M_bin11
409 prop_binSR_electron2017_bin19_TVX
410 prop_binCRTT_electron2018_bin16
411 prop_binCRTT_muon2017_bin10
412 prop_binSR_electron2017_bin12_VBS_SSWW_SM
413 prop_binSR_muon2016M_bin18_TVX
414 prop_binCRWS_muon2016M_bin17
415 prop_binCRTT_muon2016M_bin9
416 prop_binSR_electron2017_bin19_WpWpJJ_QCD
417 prop_binCRTT_muon2017_bin3
418 prop_binCRTT_electron2017_bin11
419 prop_binSR_electron2016M_bin16_TVX
420 prop_binCRTT_muon2017_bin11
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
421 prop_binCRWS_muon2016M_bin10_Fake
422 prop_binCRTT_muon2017_bin9
423 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
424 prop_binSR_muon2017_bin13_WpWpJJ_QCD
425 prop_binCRTT_electron2016M_bin13
426 prop_binCRWS_muon2016M_bin18
427 prop_binCRTT_electron2017_bin6
428 prop_binSR_electron2017_bin14_WrongSign
429 prop_binSR_electron2017_bin11_VBS_SSWW_SM
430 lep_2018
431 prop_binCRTT_electron2017_bin17
432 prop_binCRTT_electron2016M_bin19
433 prop_binCRWS_electron2017_bin16
434 prop_binCRTT_electron2017_bin7
435 prop_binCRWS_muon2018_bin13_Fake
436 prop_binCRWS_electron2017_bin17
437 lep_2016M
438 prop_binSR_muon2017_bin19_Triboson
439 prop_binCRTT_electron2016M_bin3
440 prop_binCRTT_muon2017_bin12
441 prop_binCRWS_electron2018_bin16_TTTo2L2Nu
442 prop_binSR_electron2017_bin14_WpWpJJ_QCD
443 prop_binCRWS_electron2016M_bin11
444 prop_binCRTT_muon2017_bin19
445 prop_binCRTT_muon2017_bin13
446 prop_binCRTT_electron2017_bin10
447 prop_binCRTT_electron2016M_bin4
448 prop_binSR_electron2017_bin12_WrongSign
449 prop_binCRTT_electron2018_bin10
450 prop_binSR_electron2017_bin12_WpWpJJ_QCD
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

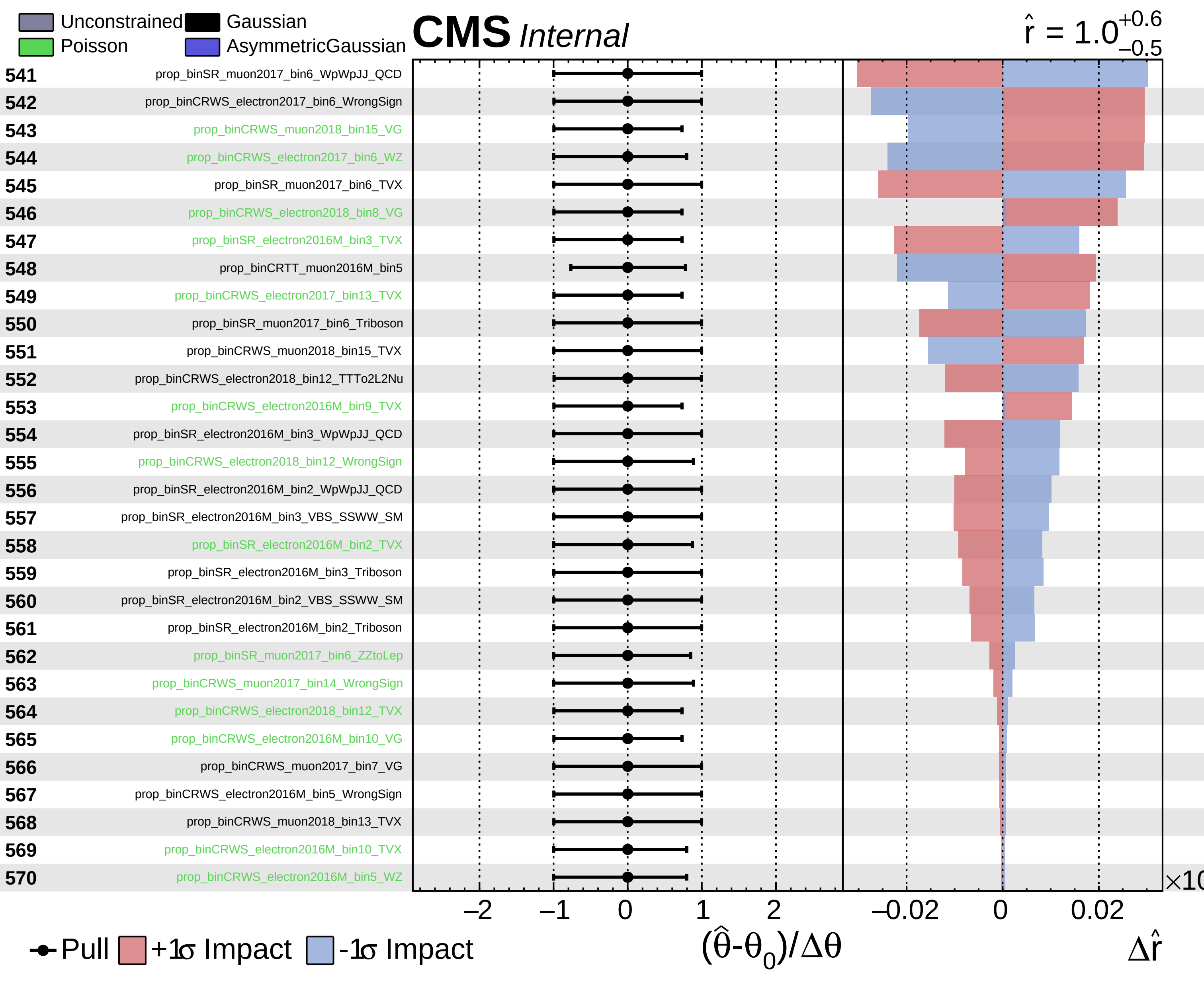


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
451 prop_binCRTT_muon2016M_bin15
452 prop_binSR_muon2016M_bin18_WrongSign
453 prop_binCRWS_electron2016M_bin16
454 prop_binCRTT_electron2018_bin4
455 prop_binSR_muon2016M_bin18_Triboson
456 prop_binCRWS_muon2017_bin7_WrongSign
457 prop_binCRTT_muon2017_bin5
458 prop_binSR_electron2017_bin19_WrongSign
459 prop_binSR_muon2017_bin6_WrongSign
460 prop_binCRWS_muon2017_bin11_TTTo2L2Nu
461 prop_binCRWS_electron2017_bin11_WrongSign
462 prop_binSR_electron2016M_bin16_WrongSign
463 prop_binCRWS_muon2017_bin13_WrongSign
464 prop_binCRTT_electron2017_bin13
465 prop_binSR_muon2017_bin6_TTTo2L2Nu
466 prop_binSR_electron2017_bin14_Triboson
467 prop_binSR_electron2016M_bin16_VBS_SSWW_SM
468 prop_binCRTT_muon2018_bin7
469 prop_binCRTT_electron2018_bin11
470 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
471 prop_binSR_electron2017_bin19_Triboson
472 prop_binCRWS_electron2016M_bin10_WrongSign
473 prop_binCRTT_muon2016M_bin7
474 prop_binSR_electron2016M_bin2_WZ
475 prop_binCRWS_muon2018_bin5_WrongSign
476 prop_binCRTT_electron2016M_bin9
477 prop_binSR_electron2017_bin11_WpWpJJ_QCD
478 prop_binSR_electron2017_bin11_TVX
479 prop_binCRWS_electron2016M_bin8
480 prop_binCRWS_electron2017_bin14_WrongSign
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

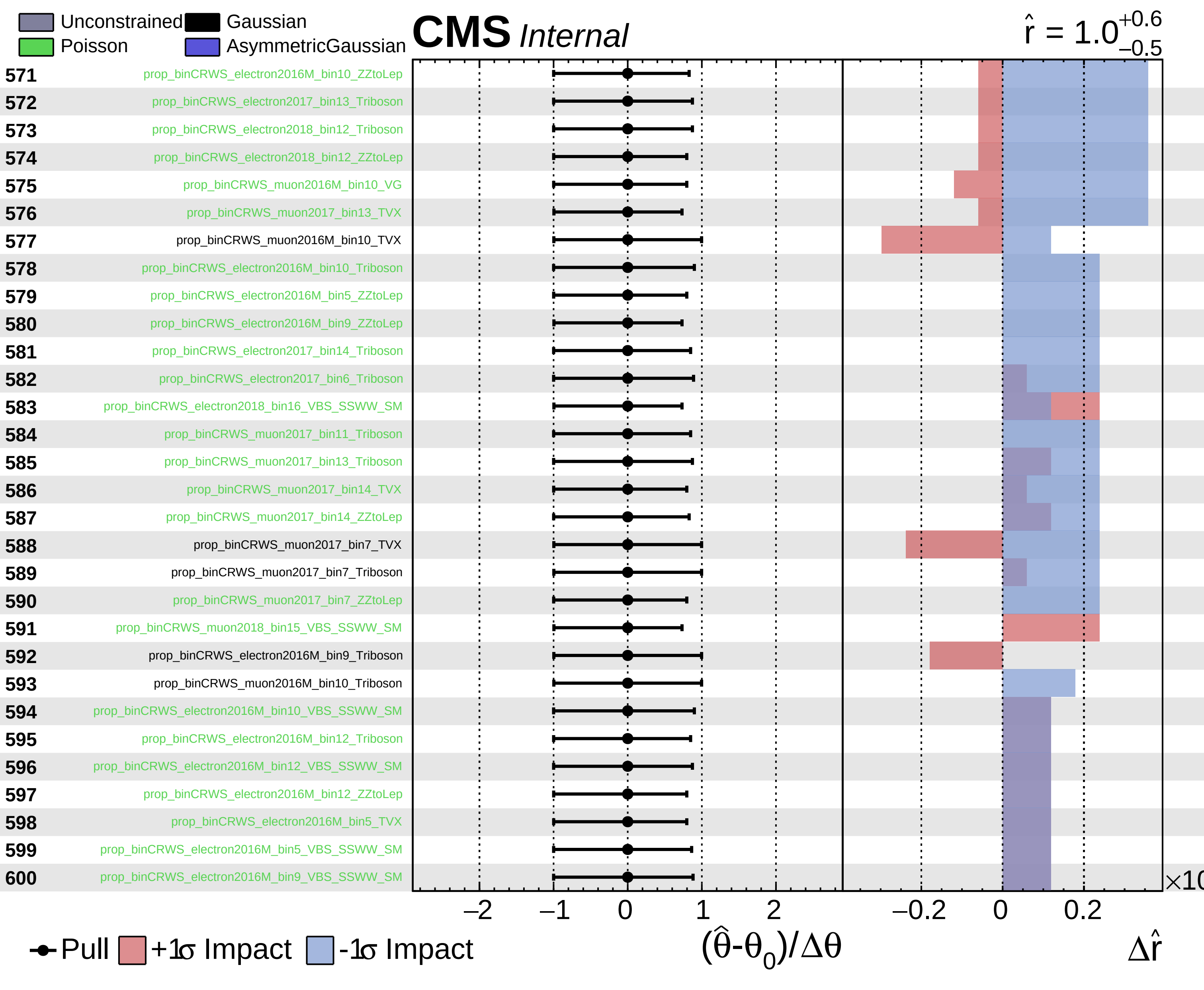

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
481 prop_binCRTT_electron2017_bin12
482 prop_binCRWS_muon2018_bin15_Fake
483 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
484 prop_binCRWS_electron2016M_bin9_Fake
485 prop_binCRTT_electron2018_bin5
486 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
487 prop_binCRWS_muon2017_bin19
488 prop_binSR_electron2017_bin12_TVX
489 prop_binCRTT_muon2016M_bin14
490 prop_binSR_electron2016M_bin3_VG
491 prop_binCRTT_electron2018_bin9
492 prop_binSR_electron2016M_bin2_WrongSign
493 prop_binCRWS_electron2016M_bin14
494 prop_binSR_electron2017_bin12_Triboson
495 prop_binSR_electron2016M_bin2_TTTo2L2Nu
496 prop_binCRWS_electron2016M_bin12_Fake
497 prop_binSR_electron2016M_bin3_TTTo2L2Nu
498 prop_binCRWS_electron2017_bin13_Fake
499 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
500 prop_binSR_electron2016M_bin16_Triboson
501 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
502 prop_binCRWS_muon2018_bin13_WrongSign
503 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
504 prop_binCRWS_muon2017_bin18
505 prop_binCRWS_muon2017_bin15
506 prop_binCRWS_muon2017_bin12_Fake
507 lep_2017
508 prop_binCRWS_muon2016M_bin10_WrongSign
509 prop_binCRTT_electron2016M_bin5
510 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



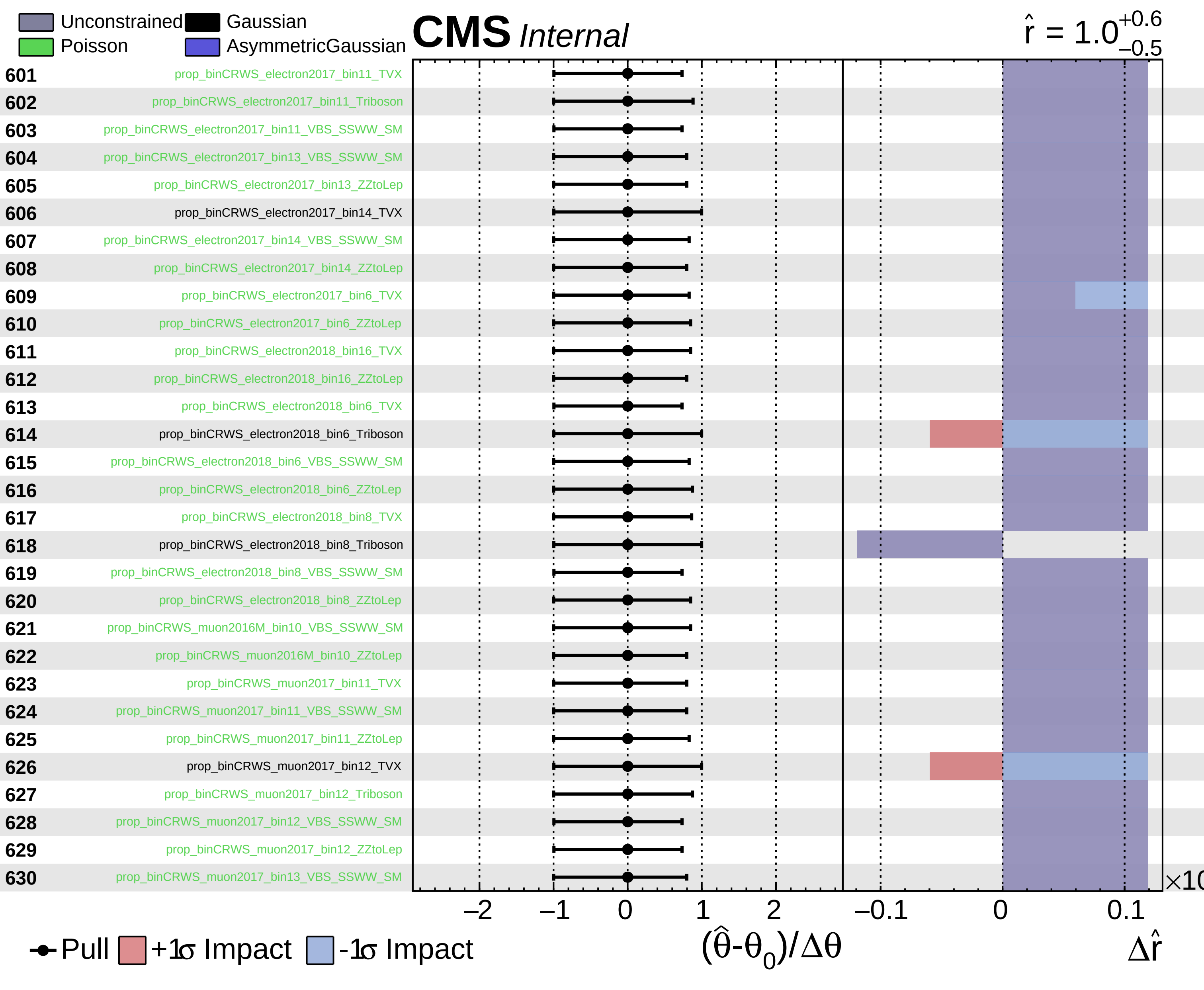
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
511 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
512 prop_binCRWS_muon2017_bin12_WrongSign
513 prop_binCRWS_electron2016M_bin5_Fake
514 prop_binCRWS_electron2016M_bin13
515 prop_binSR_electron2016M_bin3_ZZtoLep
516 prop_binCRWS_electron2016M_bin18
517 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
518 prop_binCRWS_muon2018_bin15_WrongSign
519 prop_binSR_electron2017_bin14_ZZtoLep
520 prop_binSR_electron2017_bin19_ZZtoLep
521 prop_binCRWS_electron2018_bin6_WrongSign
522 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
523 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
524 prop_binSR_electron2016M_bin2_VG
525 prop_binCRWS_electron2017_bin13_WrongSign
526 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
527 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
528 prop_binCRWS_electron2018_bin16_VG
529 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
530 prop_binSR_electron2016M_bin2_ZZtoLep
531 prop_binCRWS_electron2018_bin8_WrongSign
532 prop_binSR_muon2017_bin6_VBS_SSWW_SM
533 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
534 prop_binCRWS_electron2016M_bin12_WrongSign
535 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
536 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
537 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
538 prop_binCRWS_electron2016M_bin9_WrongSign
539 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
540 prop_binSR_electron2016M_bin3_WrongSign
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



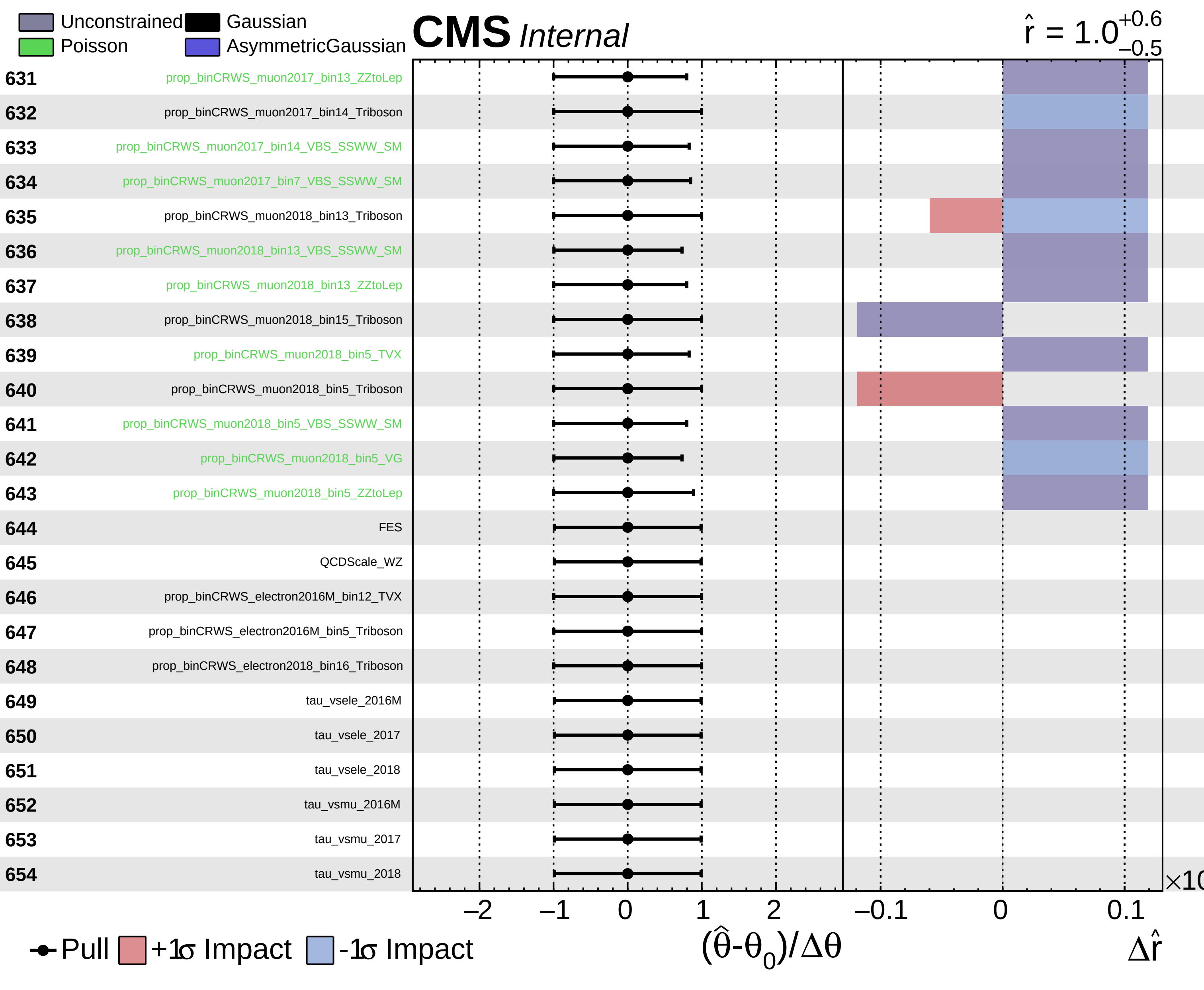
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
541 prop_binSR_muon2017_bin6_WpWpJJ_QCD
542 prop_binCRWS_electron2017_bin6_WrongSign
543 prop_binCRWS_muon2018_bin15_VG
544 prop_binCRWS_electron2017_bin6_WZ
545 prop_binSR_muon2017_bin6_TVX
546 prop_binCRWS_electron2018_bin8_VG
547 prop_binSR_electron2016M_bin3_TVX
548 prop_binCRTT_muon2016M_bin5
549 prop_binCRWS_electron2017_bin13_TVX
550 prop_binSR_muon2017_bin6_Triboson
551 prop_binCRWS_muon2018_bin15_TVX
552 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
553 prop_binCRWS_electron2016M_bin9_TVX
554 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
555 prop_binCRWS_electron2018_bin12_WrongSign
556 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
557 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
558 prop_binSR_electron2016M_bin2_TVX
559 prop_binSR_electron2016M_bin3_Triboson
560 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
561 prop_binSR_electron2016M_bin2_Triboson
562 prop_binSR_muon2017_bin6_ZZtoLep
563 prop_binCRWS_muon2017_bin14_WrongSign
564 prop_binCRWS_electron2018_bin12_TVX
565 prop_binCRWS_electron2016M_bin10_VG
566 prop_binCRWS_muon2017_bin7_VG
567 prop_binCRWS_electron2016M_bin5_WrongSign
568 prop_binCRWS_muon2018_bin13_TVX
569 prop_binCRWS_electron2016M_bin10_TVX
570 prop_binCRWS_electron2016M_bin5_WZ
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
571 prop_binCRWS_electron2016M_bin10_ZZtoLep
572 prop_binCRWS_electron2017_bin13_Triboson
573 prop_binCRWS_electron2018_bin12_Triboson
574 prop_binCRWS_electron2018_bin12_ZZtoLep
575 prop_binCRWS_muon2016M_bin10_VG
576 prop_binCRWS_muon2017_bin13_TVX
577 prop_binCRWS_muon2016M_bin10_TVX
578 prop_binCRWS_electron2016M_bin10_Triboson
579 prop_binCRWS_electron2016M_bin5_ZZtoLep
580 prop_binCRWS_electron2016M_bin9_ZZtoLep
581 prop_binCRWS_electron2017_bin14_Triboson
582 prop_binCRWS_electron2017_bin6_Triboson
583 prop_binCRWS_electron2018_bin16_VBS_SSWW_SM
584 prop_binCRWS_muon2017_bin11_Triboson
585 prop_binCRWS_muon2017_bin13_Triboson
586 prop_binCRWS_muon2017_bin14_TVX
587 prop_binCRWS_muon2017_bin14_ZZtoLep
588 prop_binCRWS_muon2017_bin7_TVX
589 prop_binCRWS_muon2017_bin7_Triboson
590 prop_binCRWS_muon2017_bin7_ZZtoLep
591 prop_binCRWS_muon2018_bin15_VBS_SSWW_SM
592 prop_binCRWS_electron2016M_bin9_Triboson
593 prop_binCRWS_muon2016M_bin10_Triboson
594 prop_binCRWS_electron2016M_bin10_VBS_SSWW_SM
595 prop_binCRWS_electron2016M_bin12_Triboson
596 prop_binCRWS_electron2016M_bin12_VBS_SSWW_SM
597 prop_binCRWS_electron2016M_bin12_ZZtoLep
598 prop_binCRWS_electron2016M_bin5_TVX
599 prop_binCRWS_electron2016M_bin5_VBS_SSWW_SM
600 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
601 prop_binCRWS_electron2017_bin11_TVX
602 prop_binCRWS_electron2017_bin11_Triboson
603 prop_binCRWS_electron2017_bin11_VBS_SSWW_SM
604 prop_binCRWS_electron2017_bin13_VBS_SSWW_SM
605 prop_binCRWS_electron2017_bin13_ZZtoLep
606 prop_binCRWS_electron2017_bin14_TVX
607 prop_binCRWS_electron2017_bin14_VBS_SSWW_SM
608 prop_binCRWS_electron2017_bin14_ZZtoLep
609 prop_binCRWS_electron2017_bin6_TVX
610 prop_binCRWS_electron2017_bin6_ZZtoLep
611 prop_binCRWS_electron2018_bin16_TVX
612 prop_binCRWS_electron2018_bin16_ZZtoLep
613 prop_binCRWS_electron2018_bin6_TVX
614 prop_binCRWS_electron2018_bin6_Triboson
615 prop_binCRWS_electron2018_bin6_VBS_SSWW_SM
616 prop_binCRWS_electron2018_bin6_ZZtoLep
617 prop_binCRWS_electron2018_bin8_TVX
618 prop_binCRWS_electron2018_bin8_Triboson
619 prop_binCRWS_electron2018_bin8_VBS_SSWW_SM
620 prop_binCRWS_electron2018_bin8_ZZtoLep
621 prop_binCRWS_muon2016M_bin10_VBS_SSWW_SM
622 prop_binCRWS_muon2016M_bin10_ZZtoLep
623 prop_binCRWS_muon2017_bin11_TVX
624 prop_binCRWS_muon2017_bin11_VBS_SSWW_SM
625 prop_binCRWS_muon2017_bin11_ZZtoLep
626 prop_binCRWS_muon2017_bin12_TVX
627 prop_binCRWS_muon2017_bin12_Triboson
628 prop_binCRWS_muon2017_bin12_VBS_SSWW_SM
629 prop_binCRWS_muon2017_bin12_ZZtoLep
630 prop_binCRWS_muon2017_bin13_VBS_SSWW_SM
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
631 prop_binCRWS_muon2017_bin13_ZZtoLep
632 prop_binCRWS_muon2017_bin14_Triboson
633 prop_binCRWS_muon2017_bin14_VBS_SSWW_SM
634 prop_binCRWS_muon2017_bin7_VBS_SSWW_SM
635 prop_binCRWS_muon2018_bin13_Triboson
636 prop_binCRWS_muon2018_bin13_VBS_SSWW_SM
637 prop_binCRWS_muon2018_bin13_ZZtoLep
638 prop_binCRWS_muon2018_bin15_Triboson
639 prop_binCRWS_muon2018_bin5_TVX
640 prop_binCRWS_muon2018_bin5_Triboson
641 prop_binCRWS_muon2018_bin5_VBS_SSWW_SM
642 prop_binCRWS_muon2018_bin5_VG
643 prop_binCRWS_muon2018_bin5_ZZtoLep
644 FES
645 QCDScale_WZ
646 prop_binCRWS_electron2016M_bin12_TVX
647 prop_binCRWS_electron2016M_bin5_Triboson
648 prop_binCRWS_electron2018_bin16_Triboson
649 tau_vsele_2016M
650 tau_vsele_2017
651 tau_vsele_2018
652 tau_vsmu_2016M
653 tau_vsmu_2017
654 tau_vsmu_2018
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$